Investigating Single-Handed Microgesture Scrolling Techniques

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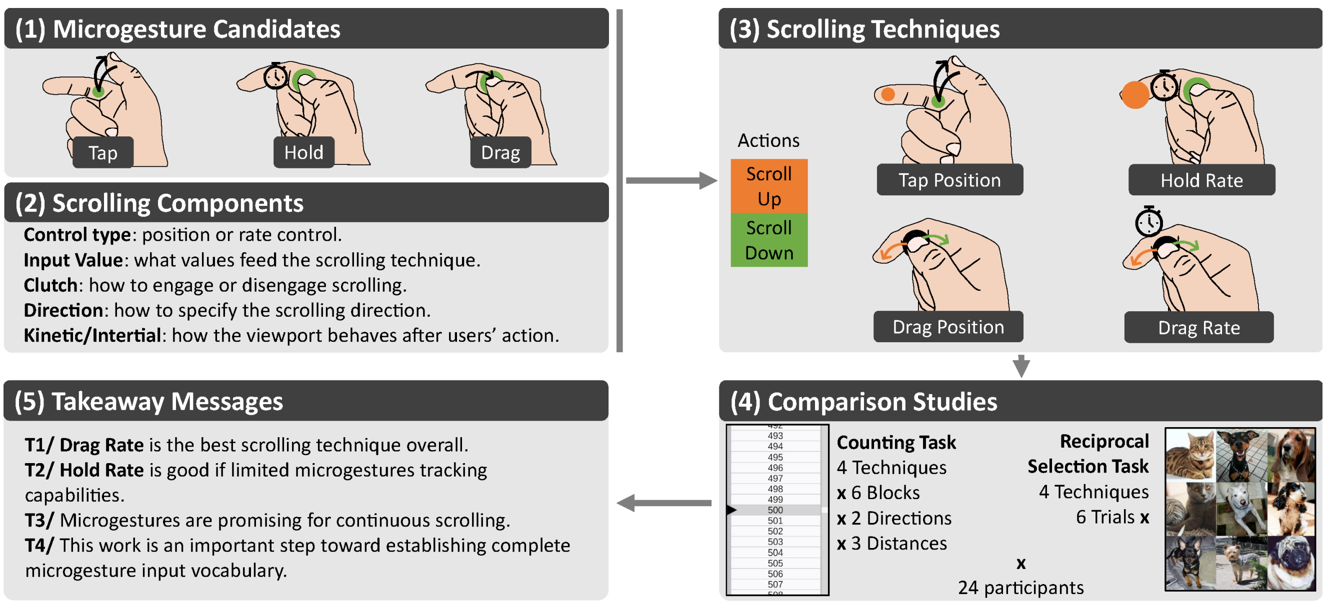

Figure 1: Visual abstract of this paper. We investigate the use of microgestures as an interaction technique for 1D scrolling. We (1) select from the literature 3 microgestures relevant for scrolling; (2) decompose scrolling into basic components; (3) instantiate 4 scrolling interaction techniques that leverage the selected microgestures, and (4) contrast them in a study with a target point and content browsing scrolling tasks; (5) We reflect on the most promising techniques and open up on future work for building a fully-fledged microgesture based interaction vocabulary

Abstract

Scrolling is ubiquitous in our daily computing experience. We explore how single-handed microgestures can be used for scrolling. Based on an analysis of the basic components necessary for scrolling, we selected 3 microgestures: Tap, Hold and Drag. Considering both rate and position controls, we designed 4 microgesture-based scrolling techniques adapted to these 3 microgestures. We contrasted these 4 techniques in a laboratory experiment with 24 participants who performed 2 tasks: a reciprocal selection task, where participants scrolled the view to reach and select a target; and a counting task, where participants scrolled the view to count image occurrences. Our results suggest that the technique based on Drag microgestures with rate control is the most effective for scrolling operations, regardless of the task. This work demonstrates that microgestures, with their advantages for frequent everyday tasks, offer a promising approach to continuous and efficient scrolling control.

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CCS Concepts

• **Human-centered computing** → **Gestural input**; **User studies**; **Laboratory experiments**.

Keywords

Microgesture, Scrolling, Interaction

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1 Introduction

Single-handed microgestures[1] are a type of gesture involving small movements of the fingers of one hand without requiring movement of the arm or wrist [13]. Typical microgestures are performed by interacting with the thumb on another finger, such as taps (i.e. tapping the thumb on a finger), holds (i.e. holding the thumb on another finger) and drags (i.e. sliding the thumb along a finger) [14]. Microgestures can also be performed by moving fingers without contact, for example flexes (i.e. bending or stretching a finger) [12]. These quick and subtle gestures allow users to interact in an eyes-free manner and perform a primary task at the same time [46]. Furthermore, since they do not involve moving the wrist or arm, microgestures do not constrain arm posture, making them less tiring than other types of gestures [12]. Their small range of motion also reduces the risk of fatigue [14, 22].

Given these advantages, microgestures are ideal candidates for common and frequent interactions such as pointing, command selection, or scrolling in various interactive contexts, including Mobile [16, 17, 27], AR/VR [6, 12], Automotive [37] and Sports [8, 45]. However, microgestures have mostly been studied as *discrete* inputs, such as command selections [8, 16, 17, 45, 46]. For tasks requiring continuous inputs, several studies combine microgestures as discrete trigger gestures with another modality that supports continuous inputs, such as gaze or arm gestures [6, 12, 16, 17]. Two technologies for sensing microgestures [47, 52] have been proposed, in which the authors illustrate the capabilities of the proposed solutions with interactive scenario including continuous variable control, but without evaluating them in experiments. The only exception of continuous input using microgestures that whose design is formally described, and benefits evaluated, is the recent study by Kim *et al.* [27] that explored *continuous* microgestures, but limited to target acquisition.

Notably, no study to date has experimentally examined the use of microgestures for continuous scrolling, i.e., the action of moving a viewport displaying text or graphic content along a given axis. Yet, scrolling is a common interaction found in almost all consumer devices, and due to its widespread use, it is an important area to study.

Our goal is to explore microgesture-based scrolling techniques that can be integrated into a complete microgesture set (see visual abstract Figure 1). Until now, scrolling has been investigated with different types of inputs (touchscreen [3, 5, 40], trackpads [3, 32], Mouse wheels [15, 39], Joysticks [50], gesture [19, 34, 36, 43], gaze [42], pen [2, 38], sound [26]) and various contexts (computer [1, 3, 4, 32, 42], mobile [3, 5, 20, 40], AR/VR [19]). Scrolling with mainstream devices such as touchpads, mouse, joysticks and smartphones is already well established and extensively explored [1–3, 5, 20, 23, 32, 39, 40, 50]. However, scrolling has rarely been studied in emerging contexts, particularly for freehand interactions such as those in AR/VR or for interacting with wearable devices. The only exception lies in Fashimpaur *et al.* who investigated scrolling in VR using wrist deflections [19]. The resulting scrolling technique relies on wrist movements, which can be tiring, especially at the high flexion angles [25] required for fast scrolling, whereas microgestures involve no wrist or arm movement.

To explore how microgestures can be used for continuous scrolling, we first analyzed scrolling from an interactive control perspective to identify its essential *components*, that is, the minimal functional requirements necessary to implement scrolling behavior. Second, we selected a subset of microgestures whose morphological *characteristics* made them particularly suitable for fulfilling these components. Finally, we combined this design space of scrolling *components* and microgesture *characterstics* with a rate or position *control*, to create 4 microgesture-based scrolling techniques that we contrast in a laboratory experiment with 24 participants asked to perform 2 tasks: a *reciprocal selection* task, where participants scrolled the view to reach and select a target; and a *counting* task, where participants scrolled the view to count image occurrences. Our results suggest that techniques that use rate control, particularly the technique based on Drag microgestures with rate control, are the most effective for scrolling operations, regardless of the task.

This paper makes the following contributions: 1) Four novel scrolling techniques involving three types of microgestures Tap, Hold and Drag, including two techniques based on rate control and two based on position control. 2) An empirical study comparing the four scrolling techniques, highlighting their relative strengths and weaknesses and experimentally demonstrating for the first time the viability of microgestures for scrolling. 3) Design insights for scrolling techniques based solely on microgestures, with scrolling representing an important step toward establishing microgestures as a fully-fledged interaction input.

2 Related Work

2.1 Microgesture Interactions

Microgestures define an emerging form of hand gesture interaction, relying on subtle movements of fingers performed without wrist or arm movements [13]. Microgestures can be performed eyes-free, typically to interact while performing a primary task [46]. In the very active field of hand microgestures, the literature on free-hand and grasping microgestures [9] explores a rich set of microgestures. This richness is achieved by considering fine characteristics of a microgesture. A microgesture is commonly characterized by its action (physical motion such as a tap or a swipe), by its actuator (the finger performing the action), and its receiver (the surface touched during the action, which is very often a part of the hand, but can also be an object held in the hand). Chaffangeon Caillet *et al.* [13] presented

[1] In the following we use the term "microgesture" to refer to a single-handed microgesture

µGlyph, a notation for precisely describing microgestures and their characteristics. In µGlyph, a microgesture is described using 4 key components: movement, context, actuator, and surface. The *movement* can be a flexion (▼), an extension (▲), an abduction (◄), an adduction (►) or be absent (■). The *context* specifies whether the movement occurs in the air (○○), in contact with a surface (●●), or whether it occurs on (○●) or off (●○) a surface. The combination of *movement* and *context* (the action) allows for higher-level descriptions such as touch ($\overset{\circ\bullet}{\blacktriangledown}$), hold ($\overset{\bullet\bullet}{\blacksquare}$) and release ($\overset{\bullet\circ}{\blacktriangle}$). The *actuator* is indicated as a subscript of the *movement* glyph, for example ${}_{t}\overset{\circ\bullet}{\blacktriangledown}$ for a touch of the thumb. Finally, the *surface* (the receiver) is denoted by the contact glyph (●) with an associated index; for example ${}_{t}\overset{\circ\bullet}{\blacktriangledown}\left({}_{i}\bullet\right)$ represents a touch of the thumb on the index finger. Throughout the paper, we use the µGlyph notation to describe microgestures. Taking advantage of this rich set of microgestures, studies have examined interactions based on microgestures in various contexts.

For interaction, microgestures have mostly been studied as discrete inputs, for instance for command activations [8, 37, 45, 46]. Boldu *et al.* studied discrete thumb-to-finger tap, swipe-down, and swipe-left microgestures to control a music player or a stop watch while running [8]. Neßelrath *et al.* studied discrete stretch microgestures performed by fingers of the two hands on the steering wheel to trigger in-car controls (e.g. turning lights and windows on and off) [37]. Tan *et al.* studied discrete microgestures including thumb-to-finger taps and finger taps on a bike structure for input controls (e.g. headlight control) while biking [45]. Wambecke *et al.* studied discrete thumb-to-finger tap microgestures (taps of the thumb on the upper/lower phalanx of the index and middle fingers) for navigating a two-level menu [46].

Microgestures have also be used for tasks requiring continuous inputs. For those tasks, several studies combine microgestures as discrete trigger gestures with another modality that supports continuous inputs, such as arm gestures, gaze and touch [6, 12, 16, 17]. Bailly *et al.* designed a technique for controlling a drone using a world in miniature of the environment [6]. Continuous interactions such as drone translation and rotation are performed by wrist rotation or hand translation. Flex microgestures are used to trigger a +/-90-degree rotation of the world in miniature. Chaffangeon Caillet *et al.* proposed a two-step 3D selection technique in AR/VR [12]: the initial continuous selection is handled with gaze, and discrete thumb-to-finger tap microgestures are used to navigate among the items selected by the gaze. Faisandaz *et al.* designed a technique for people with visual impairment to explore an audio-tactile map [16]. Continuous interaction is performed by touch interaction on the map, while discrete microgestures are used to trigger audio feedback about the selected point on the map. Similarly, Faisandaz *et al.* designed an eyes-free text selection technique [17] based on the combination of touch and microgestures.

Nevertheless, continuous interaction with microgestures is supported by sensing technologies. For instance technologies based on a touch-sensitive glove [47] or visual tracking [27, 52] are capable of detecting the continuous touch position of the thumb along the fingers. In studies devoted to sensing technologies, authors illustrate the capabilities of their sensing technologies either using envisioned scenarios [47] or demonstrations [52]. For instance, Zoppis *et al.* presented a demo application involving continuous microgestures: the task consists of editing a 3D point cloud and sliding the thumb on the index finger allows the radius of the cursor (a sphere) to be continuously adjusted [52]. However, this technique is not evaluated in experiments and is used solely to demonstrate the proposed detection technology based on visual tracking. To our knowledge, only Kim *et al.* ś recent study has focused on the design and evaluation of pointing techniques based on thumb-to-index motion in AR/VR [27]. This study highlights the potential of microgestures as full continuous input and motivated us to explore continuous scrolling with microgestures.

2.2 Scrolling

Scrolling is a common task on all consumer devices, such as smartphones and computers. Conceptually, it consists of controlling the vertical or horizontal displacement of content relative to a fixed viewport, typically to navigate through information that extends beyond the displayed area. One example is vertical scrolling of a list, which involves moving the list along the 1D vertical axis. List scrolling has been used and studied on many input devices, including touchscreen [3, 5, 40], trackpads [3, 32], mouse wheels [15, 39], joysticks [50], gestures [19], gaze [42], pen [2] and sound [26].

2.2.1 Scrolling control types. Three types of control [49] for scrolling have been characterized: position, rate and acceleration controls.

Position control consists in directly controlling the content position. The input specifies the operation to be applied to the scrolling position. This type of control requires a clutch mechanism (i.e., a distinct way to engage and disengage scrolling) for flick-based techniques [5]. Experimental evidence suggests that it is particularly well suited to tasks involving small scrollable content [28].

Conversely, *Rate* control consists of indirectly controlling the content position by setting an autoscroll speed. More precisely, the input directly specifies the scrolling speed. The use of isometric or elastic devices (i.e., self-centering mechanisms) is particularly well suited to this control [10]. Unlike Position control, Rate control is recommended for tasks involving large scrollable content [28]. With a continuous speed, this type of control generally requires low-effort for long scrolling distances.

The third one, *acceleration* control, consists of indirectly controlling the content position by adjusting the autoscroll speed. More precisely, the input specifies whether to increase or decrease the current scrolling speed. Compared to the other two forms of control, this approach is generally more difficult to master and less stable [35].

2.2.2 Gesture Scrolling. Gestures are a widely used input modality for scrolling, ranging from everyday swipes on mobile devices to mid-air interactions.

In daily use, the most common in-contact gestures are simple drags or swipes on tactile surfaces. These gestures directly control position on tactile device [3, 5, 40] or trackpads [3, 32]. More dynamic gestures, such as flicks, trigger inertial scrolling by simulating a quick release [2, 5]. Beyond these, more complex in-contact gestures have also been explored for scrolling, such as radial gestures, where circular finger motions are used to navigate through content [34, 36, 43].

Many mid-air gesture scrolling techniques have been studied and incorporated into commercial devices, including AR/VR devices. Indeed, Meta Quest and Apple Vision Pro devices allow the users to scroll using pinch gestures, then moving their hand on a vertical axis. In addition, they enable freehand pointing interaction, allowing the user to grab the scroll bar and manipulate it. Even if these gestural scrolling techniques exist and are effective for occasional interactions, the gorilla arm effect can still occur during long-term interactions [22].

To avoid the "gorilla arm" effect, some techniques have been studied to prevent arm movements. Fashimpaur *et al.* [19] proposed techniques for both position and rate control. These techniques use the wrist angle as an input value, scrolling as the angle changes.

Currently, only one paper deals with the use of microgestures for scrolling. Kin *et al.* [29], at the end of their paper, mention the possibility of using discrete microgestures for discrete scrolling. They discuss the use of discrete Swipe on the index to perform discrete scrolling, in order to navigate a Netflix-like application. Although scrolling is an essential continuous interaction, no study has yet focused on scrolling techniques based solely on microgestures.

3 Design

3.1 Scrolling Components

In order to design efficient scrolling interaction techniques relying on microgestures, we first need to carefully characterize the interaction of scrolling itself. Only then, will we be able to reflect on which microgestures are best suited.

Despite the extensive research on scrolling, as reviewed in Section 2.2, we are not aware of any formal characterization. To address this, we propose a characterization informed by both existing literature and commercial scrolling techniques, that divides scrolling into components that can efficiently describe the interaction. It is intended as a practical tool to support both our own design process and the broader community in designing scrolling techniques.

The characterization consists of five key components that capture the essential aspects of existing techniques: **Input Type**, **Direction**, **Clutch**, **Control type** and **Kinetic/Intertial**.

Input Type. refers to the type of raw values provided by the input modality (mouse, touch, gaze, etc.). These values can be discrete or continuous. For example, a touchscreen can interpret finger or stylus positions to provide continuous values [2, 5, 40]. Buttons, on the other hand, typically only provide discrete values (pressed or not) [29]. Discrete inputs may emulate continuous values through additional interpretation, for instance by accumulating time (e.g., holding down a keyboard direction key or a game controller button). However, user's control over these emulated continuous values remains limited. The **Input Type** component is mandatory for scrolling.

Direction. defines how the input operations are mapped to a certain directional scrolling (typically, upward/downward in the case of 1D vertical scrolling). In techniques that rely on continuous **Input Type**, the scrolling direction is typically inferred from the evolution of the input value. This can be done by monitoring successive changes [2, 5, 19, 32, 50] (e.g., moving a finger upward will scroll downward), or by mapping the range of possible values to different directions [3, 19, 50] (e.g. scrolling upward when the finger is in the top half of the screen). In contrast, techniques that use discrete **Input Type** usually derive direction from multiple input sources (e.g. different keyboard keys or game controller buttons). The **Direction** component is mandatory for 1D scrolling to be bidirectional.

Clutch. refers to how users can temporarily disengage the scrolling operations [48], to facilitate users decisions to trigger and stop scrolling when desired. In typical touch-based scrolling, **Clutch** is achieved by lifting the finger from the surface [2, 5, 32]. In the case of a joystick-based scrolling, **Clutch** can be achieved by returning the joystick to its default (neutral) position [50]. With directional pads, **Clutch** is explicitly achieved by not pressing the buttons. If **Clutch** cannot be explicitly controlled by the user, it may result on unintended scrolling actions or limited control. Therefore, the **Clutch** component is essential to ensure usable scrolling.

Control type. describes how **Input Type** is mapped to the way scrolling is controlled. Scrolling techniques are frequently characterized by their **Control type** [49]: *Position* (or zero order), *Rate* (or first order), and *Acceleration* (or second order)[2] control. In the case of scrolling, *Position* and *Rate* are typically used. *Position* means that value changes of the **Input Type** are mapped to scrolling, and conversely, that if it does not change then it does not scroll. *Rate* means that the **Input Type** value is mapped to scrolling, typically each **Input Type** value is mapped to a specific scrolling velocity. **Control type** is therefore constrained by both the **Input Type** and **Clutch** components. For example, an input device that provides a continuous **Input Type** but clutches when its value is reset (e.g., joysticks) is not adapted to position control. Indeed, tilting the joystick to a certain angle and resetting its position corresponds to a back-and-forth motion whose net displacement is zero. Similarly, an input device generating only discrete **Input Type** may support position, but will be less suited for rate control as it readily provides only a binary control of the value. **Control type** is a mandatory component present in all scrolling techniques.

Kinetic/Intertial. corresponds to a kinetic (or inertial) scrolling behavior that occurs when users disengage from scrolling [40] intended to facilitate interaction when **Clutch** is frequently required. In existing scrolling techniques, kinetic scrolling is typically used with position control techniques [2, 5, 32]. Typically, in touch-based scrolling, users can "throw" the content to reduce the number of sequential inputs required to travel a certain distance. **Kinetic/Intertial** is optional in scrolling techniques, but remains particularly relevant for some position control techniques.

As a summary, five key components characterize scrolling techniques. Three of these, **Input Type**, **Direction**, and **Clutch**, are mandatory and theoretically independent. A fourth component, **Control type**, is essential and constrained by both **Input Type** and **Clutch**. The fifth, **Kinetic/Intertial**, is optional and best suited to specific scrolling techniques.

[2] As seen in the related work, Acceleration control approaches should generally be avoided [35]. Note also that "Hybrid" control techniques combine rate and position control [11].

	Input Type	Directions	Clutch	Control Type	Considered for experiment
Tap $_t$▼($_i$●); $_t$▲	**Discrete** *Contact or not*	**2** *Phalanx*	**Yes** *Contact*	**Position**	**Yes** $\text{Tap}_{Position}$
Hold $_t$▼($_i$●); $_t$■	**Continuous** *holding time*	**2** *Phalanx*	**Yes** *Contact*	**Rate**	**Yes** Hold_{Rate}
Force Press $_t$▼ ($_i$●) or $_t$▲ ($_i$●) *lo* → *hi* *hi* → *lo*	**Continuous** *Thumb normal force*	**2** *Phalanx/Force*	**Yes** *Contact/Force*	**Rate**	**No** Due to **Input Value**
Drag $_t$▼($_i$●) or $_t$▲($_i$●)	**Continuous** *Drag position*	**2** *Direction*	**Yes** *Contact*	**Position Rate**	**Yes** $\text{Drag}_{Position}$/Drag_{Rate}

Table 1: Mapping microgestures to minimal scrolling components. Green cells are direct bindings, orange cells when a tweak is required, and red cells no matching is possible.

3.2 Designing scrolling techniques leveraging microgestures

Single-handed microgestures are diverse and several of them could, in principle, support scrolling interaction. To narrow the scope of this investigation, we consider the following aspects.

First, when available, scrolling operations are relatively frequent and as thus must rely on a motor input action that is easy to perform and not tiring. While microgestures are generally easy to perform, some may require greater dexterity and may consequently induce fatigue more rapidly. We therefore deliberately exclude microgestures that may necessitate substantial dexterity such as finger flexion, extension, abduction and adduction. This constraint also excludes gestures involving wrist flexion, which is known to induce fatigue [7], such as Fashimpaur *et al.*'s technique [19].

Second, various sensing technologies with differing levels of precision have been developed to detect single-handed microgestures (e.g., Apple Vision Pro, Meta Quest, and prior academic research [21, 44, 47]). Most of these focus on thumb-to-finger microgestures, such as detecting finger contact, the location of that contact and the pressure applied. For this reason, we exclude gestures involving finger flexion or bending (e.g., flexion, extension, abduction, and adduction), as these gestures are less commonly supported by current sensing technologies. Our focus is therefore on thumb-to-finger microgestures.

As a result, our investigation focuses on the Tap, Hold, Force Press, and Drag microgestures. Table 1 presents how these microgestures can accommodate the previously described components (**Input Type**, **Direction** and **Clutch**) to control scrolling. In the following, we present the *scrolling interaction techniques* designed for the most promising microgestures and subsequently describe the iterative refinement process used to define the transfer function [39, 40] of each technique.

3.2.1 *Tap.*

The Tap microgesture corresponds to a simple "tap" of the thumb on another finger. $_t$▼($_i$●); $_t$▲ – The thumb touches then release the index. By default it offers a single discrete **Input Type** variable (i.e., in-contact or not), which is insufficient to support efficient 1D-scrolling that also requires to specify a **Direction**, at least. This can be easily achieved by mapping two different phalanxes of the finger to different scrolling directions. The binary nature of Tap makes **Clutch** easy as, by default, it scrolls only in response to a Tap from the users. However, this binary nature requires adjustments to enable control over **Input Types**, which can be achieved through the scrolling transfer function (see next paragraph), for example by considering consecutive taps to increase the scrolling distance.

$\text{Tap}_{Position}$. This scrolling technique relies on the Tap microgesture and *position control* (see Figure 2). It is a discrete scrolling technique (akin to Kin *et al.* [29]) that scrolls the view of a distance S_d for each Tap performed. Notably, like the input, the resulting scroll displacement is discrete, with the viewport jumping by the distance S_d. When users perform Taps on the distal or proximal phalanx, the viewport moves upward or downward, respectively. If users perform consecutive Taps on the same phalanx, the scrolling distance increases proportionally as the time interval t between taps decreases.

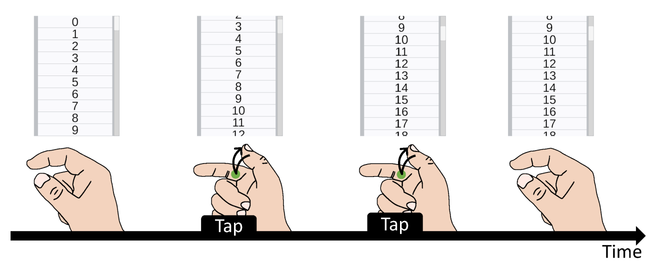

Figure 2: $\text{Tap}_{Position}$ technique: Users taps their thumb on a phalanx to scroll. Quick repeated taps scroll faster.

The transfer function of $\text{Tap}_{Position}$ maps consecutive taps in short intervals to larger scrolls, and conversely long intervals to shorter scrolls. To this end, we use an inverted exponential curve as the initial basis, given its suitability for modelling this mapping. We then adapt this curve and introduce several additional parameters to more closely align it with the intended behavior, resulting in the following equation 1. In this equation, parameter a defines the longest interval associated with the shorter scrolling distance c, parameter b controls the gain of the curve. Through iterative refinement (see Section 3.2.5), the parameters were set to $a = 0.8$s, which enables short distances to be triggered without excessive delay, as well as $b = 6$ and $c = 30$. The distance SU_d is expressed in unity values, and $S_d = SU_d/45.8$ in centimeters.

$$SU_d = e^{(a - \min(t,a)) \cdot b} \cdot c \tag{1}$$

3.2.2 Hold. The Hold microgesture corresponds to prolonged contact between the thumb and the finger. $_{t}\overset{\circ\bullet}{\blacktriangledown}(_{i}\bullet)$; $_{t}\overset{\bullet\bullet}{\blacksquare}$ – The thumb touches the index finger and maintains contact. As such, it provides a theoretically unlimited continuous input (contact duration). However, the contact time can only increase, and the transfer function must determine how to convert it to **Input Type**. As with Tap, different phalanxes can be used for the **Direction** choice. Finally, **Clutch** can be performed at any time by simply releasing contact between the thumb and finger.

Hold$_{Rate}$. This scrolling technique relies on the Hold microgesture and *rate control* (see Figure 3). When users perform a Hold on the distal or proximal phalanx, the interface scrolls upward or downward, respectively. Scrolling speed increases with Hold duration until it reaches a maximum value. The initial low-speed phase allows for fine adjustments, while the higher-speed phases make facilitate access to distant targets. Scrolling stops as soon as fingers are not in contact anymore.

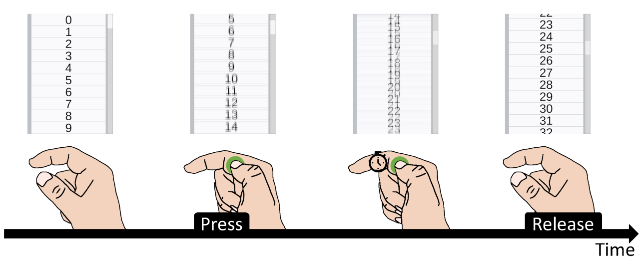

Figure 3: Hold$_{Rate}$ technique: Users holds their thumb on a phalanx to scroll. Longer holds scroll faster.

The transfer function of Hold$_{Rate}$ enables slow scrolling for fine adjustments at first, before to progressively transition to higher speeds for browsing long contents, with a cap to minimize overshoots. While Hold is held, the view scrolls by a distance S_d every 20 milliseconds, depending on the duration t of the Hold. We use a sigmoid curve as a base, since it readily offers three phases: an initial phase with a slow slope, a middle phase with a higher slope, and a final phase with a slow slope. This is frequently used for scrolling transfer functions [3, 39]. This design choice led us to equation 2. In this equation, parameter a specifies the onset of the acceleration phase (i.e., it roughly indicates the time t at which acceleration begins), while parameter b determines the maximum speed reached. Through iterative refinement (see Section 3.2.5), the parameters were set to $a = 0.4$, corresponding to a short delay before acceleration, and $b = 4000$, enabling a high speed suitable for browsing long content. The distance SU_d is expressed in unity values, and $S_d = SU_d/45.8$ in centimeters.

$$SU_d = \frac{b}{1 + e^{-\frac{t}{a} + 4}} \tag{2}$$

3.2.3 Force Press. The Force Press microgesture corresponds to a continuous contact between the thumb and finger, and features additional force-sensing capabilities. $\underset{lo \rightarrow hi}{_{t}\overset{\bullet}{\blacktriangledown}}\ (_{i}\bullet)$ or $\underset{hi \rightarrow lo}{_{t}\overset{\bullet}{\blacktriangle}}\ (_{i}\bullet)$ – The thumb increases or decreases the normal force on the index. As such, it can provide a bounded continuous **Input Type** based on the amount of force applied to the thumb and converted via a transfer function. Once again, **Direction** can be specified by the phalanx like Tap and Hold. The **Clutch** requirement is met by simply segmenting the input into contact or no contact.

However, two problems are faced in practice. First, while some optical sensing technologies can infer force variation in microgestures to some extent [21], it is inherently difficult to do accurately due to optical occlusion. Second, reducing force is also challenging, as previous work has shown that force input is unidirectional [51], suggesting that it is easier control an increase of force than a decrease.

We thus exclude the Force Press microgesture from our study due to the technical limitations in tracking and human inability to control pressure.

3.2.4 Drag. The Drag microgesture corresponds to displacements of the thumb on the surface of another finger (typically the index). $_{t}\overset{\bullet\bullet}{\blacktriangledown}(_{i}\bullet)$ or $_{t}\overset{\bullet\bullet}{\blacktriangle}(_{i}\bullet)$ – The thumb moves along the index finger. As such, the position of the contact point on the receiving finger provides a bounded continuous **Input Type**. **Direction** can be easily mapped in different ways, either based on the position of the contact point or the direction of its movement if moving. As Tap and Hold microgesture, **Clutch** is easily achieved by releasing contact. The ability to control the continuous **Input Type** opens up the possibility of using either *position* or *rate* control-based scrolling techniques.

Drag$_{Position}$. This technique relies on the Drag microgesture and *position control* (see Figure 4). It mimics indirect touch-based scrolling: thumb's displacement on the finger T_d are continuously mapped to scrolling distance S_d, following transfer function 3. A Drag towards the distal (respectively proximal) phalanx scrolls upward (respectively downward). Since this is position control, scrolling stops as soon as the thumb remains still on the finger, or if it releases contact unless thumb's velocity is above a certain threshold, in which case inertial scrolling is applied replicating the formula from Android Flick described in [40]. This threshold corresponds to a displacement of 5 % of the finger length within the last 100 ms. For the inertial speed I_s slowdown function we chose: $I_s = I_s * (1 - 0.04)$.

The transfer function of Drag$_{Position}$ provides different scrolling sensitivities to enable different drag speeds. To achieve this, we defined three thresholds ($t1$, $t2$, and $t3$) that segment the input

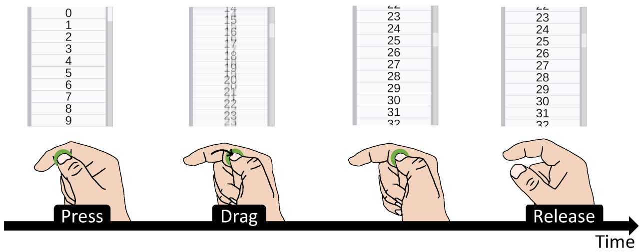

Figure 4: Drag$_{Position}$ technique: Users scrolls by moving their thumb alongside the finger. Faster moves scroll faster.

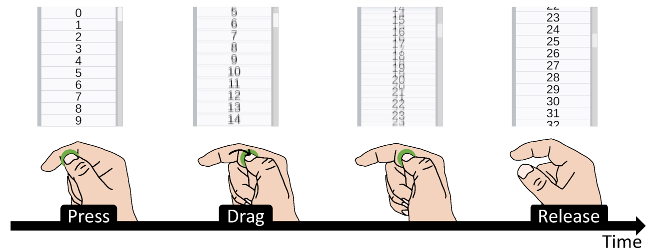

Figure 5: Drag$_{Rate}$ technique: Users move their thumb away from a central point alongside the finger. Greater distances from the central point scroll faster.

into distinct speed ranges. Each threshold interval is associated with a corresponding speed parameter ($s1$, $s2$, and $s3$), with linear interpolation between thresholds. This led us to the formulation shown in equation 3. If $T_d < t_1$, the scrolling speed is proportional to s_1. If $T_d \in [t_1, t_2[$, the speed is computed as an interpolation between s_1 and s_2. If $T_d \in [t_2, t_3[$, it is an interpolation between s_2 and s_3. If $T_d \leq t_3$, the scrolling speed is proportional to s_3. Through iterative refinement (see Section 3.2.5), the parameters were set to $t_1 = 0.005$, $t_2 = 0.04$, $t_3 = 0.2$, $s_1 = 300$, $s_2 = 6000$ and $s_3 = 9000$. The distance SU_d is expressed in unity values, and $S_d = SU_d/45.8$ in centimeters.

$$SU_d = \begin{cases} T_d \cdot s1 & \text{if } |T_d| < t1, \\ T_d \cdot \left(\dfrac{|T_d| - t1}{t2 - t1} \cdot s2\right) & \text{if } t1 \leq |T_d| < t2, \\ T_d \cdot \left(\dfrac{|T_d| - t2}{t3 - t2} \cdot s3\right) & \text{if } t2 \leq |T_d| < t3, \\ T_d \cdot s3 & \text{if } |T_d| \geq t3. \end{cases} \quad (3)$$

Drag$_{Rate}$. This technique relies on the Drag microgesture combined with *rate control* (see Figure 5) and mimics the joystick of a game controller. Absence of contact and contact near the middle of the finger (with a 10% neutral zone around the center) does not produce any scrolling. Moving the thumb toward the fingertip scrolls upward, while moving it toward the finger base scrolls downward. A small displacement from the center results in slow scrolling, and the farther the thumb moves away from the center, the faster the scrolling. While the contact is maintained outside the neutral zone, the view scrolls by a distance S_d every 20 milliseconds, depending on the distance between center of finger and thumb contact position P_d.

The transfer function of Drag$_{Rate}$ increases scrolling speed as the thumb moves farther away from the center. To achieve this, we chose a power-of-two function, which grows exponentially (equation 4). The a parameter controls how much the thumb's distance from the center influences the scrolling speed, while parameter b acts as a speed-scaling factor. Through iterative refinement (see Section 3.2.5), the parameters were set to $a = 18$ and $b = 300$, allowing the technique to support both slow and high scrolling speeds. The distance SU_d is expressed in unity values, and $S_d = SU_d/45.8$ in centimeters.

$$SU_d = 2^{P_d \cdot a - 3} \cdot b \quad (4)$$

3.2.5 Transfer function optimization. Each of the previously described scrolling transfer functions was refined through an iterative process that first involved the paper's authors and later four external participants. We began with a series of trial-and-error iterations until we reached a satisfactory preliminary set of transfer functions and parameter values. We then conducted an informal pilot study with four external participants. The goal of this pilot study was to further refine the transfer functions based on feedback from individuals not involved in the development of the techniques.

The task was similar to the Reciprocal Selection Task used in the experiment (see Section 4.2.1), but incorporated a wider range of target distances to reach a more general optimization of the transfer functions rather than tailoring them specifically to the parameters of the experiment. Participants had to scroll both upward and downward to select targets positioned at eight different distances from the starting point: 5 % of the list component (1.3 cm), 20 % of the list component (5.24 cm), 50 % of the list component (13.1 cm), 100 % of the list component (26.2 cm), 200 % of the list component (52.4 cm), 400 % of the list component (104.8 cm), 600 % of the list component (157.2 cm), and 800 % of the list component (209.6 cm). A block therefore consisted of 16 trials (2 Directions × 8 Distances).

Participants first completed one block using the most recent parameter set (either the authors' original parameters or those modified by the previous participant). After the first block for each technique, we explained the role and impact of each parameter to the participants. If they felt that the transfer function could be improved, they adjusted the parameters, evaluated the resulting technique using a sandbox list, and repeated the block with the updated parameters. We asked participants to verbalize any proposed changes, enabling us to better understand the reasoning behind those changes (e.g., one participant reported decreasing the Hold$_{Rate}$ maximum speed in order to make the content easier to perceive at high speeds). Once participants were satisfied with the parameters for a given technique, they proceeded to the next technique.

The parameter configuration that emerged after the fourth participant's session was selected as the final transfer-function parameter set. We did not observe major iterative back-and-forth adjustments across participants. Indeed, when a new participant began their session, they typically experimented with a few alternative settings before ultimately returning to the previous configuration, or made only minor adjustments. Overall, the changes from one participant to another were minimal and did not result to substantial differences in the transfer-function parameters.

4 Study: Methodology

We conducted an experiment to contrast the performance of the different micro-gesture based scrolling techniques described in section 3.2. Our goal was to evaluate and compare their performance, effectiveness and usability over different scrolling tasks and difficulties.

We tried to design different scrolling techniques, accommodating different microgestures. As a result, some rely on discrete **Input Type** ($\text{Tap}_{Position}$) and other continuous **Input Type** (Hold_{Rate}, $\text{Drag}_{Position}$ and Drag_{Rate}). Given the granularity of control that continuous **Input Type** offer, we expect the corresponding techniques to yield better performance than other techniques, overall. Finally, inertial scrolling techniques relying on 'flicking' gestures are common on everyday devices such as phones and trackpads [2, 5, 20, 40], we therefore expect users to perform well with the corresponding technique. Altogether, we have the following hypothesis:

H1: Participants will prefer and perform better with $\text{Drag}_{Position}$ and Drag_{Rate} than with the other techniques because they support continuous **Input Type**.

H2: Participants will prefer and perform better with Hold_{Rate} than with $\text{Tap}_{Position}$ because it supports partially controlled continuous **Input Type**.

H3: participants will perform better with $\text{Drag}_{Position}$ thanks to skill transfer from touch-based devices.

4.1 Procedure

Participants were welcomed by the experimenter and invited to sit at a desk, where a monitor was placed to display the experimental interfaces. After reading and signing the informed consent form, the experimenter explained the purpose and procedure of the experiment to the participants (see supplementary material for the introduction and instruction speeches). Participants were then free to ask questions, if any. Then, they were asked to answer some demographic questions (age, gender and dominant hand). At this stage, the experimenter equipped the participants' dominant hand with reflective tracking spheres (Figure 7).

Participants then began the *experiment*. For each Technique, the experimenter introduced it, followed by a two-minute familiarization phase during which participants could scroll through a numbered list using the corresponding Technique. Participants then completed 6 blocks of 6 trials of one *Experimental Task* (see section 4.2) and answered the corresponding Likert-like questions (see section 4.3), before doing the same for the other *Experimental Task*. This sequence was repeated for all remaining Techniques. The session ended with an open discussion where participants could share comments or ask questions.

4.2 Experimental Tasks

Scrolling is a common and widely used input method that supports many types of tasks, each involving different user goals and contexts. Nevertheless, scrolling interaction techniques are typically evaluated with a pointing-like task. In these tasks, participants are typically provided with information about the target's position and instructed to scroll until they locate and select it [2, 3, 15, 23, 32].

While these experiments may differ in parameters such as target distance, size, or direction, their findings have limited validity, as they primarily reveal how each technique performs under a specific scrolling behavior.

However, scrolling can also involve other behaviors, such as browsing or reading the content. To enhance the validity of scrolling experiments, some studies have evaluated scrolling techniques in tasks encouraging searching or browsing behaviors, using a randomly ordered list [5] or asking to count images in an unordered gallery [19, 20].

In this study, we contrast scrolling techniques in two Tasks, each encouraging a different scrolling behaviour: a *Reciprocal Selection Task* and a *Counting Task*.

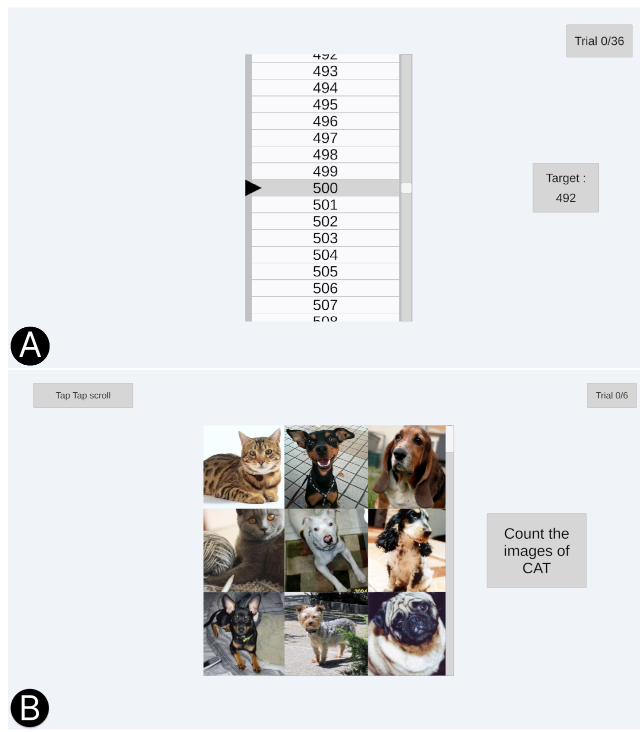

Figure 6: A) Reciprocal Selection Task Interface. B) Counting Task Interface.

4.2.1 *Reciprocal Selection task.*

The *Reciprocal Selection Task* required participants to perform sequences of target selection. To select a target, participants had to scroll a list of numbers "as quickly and accurately as possible" until they reached the target, which they had to select. The experimental interface (illustrated in Figure 6-A) displayed the list of items in the center and the target number on the right. The interface component displaying the list had a height of 26.2 cm, corresponding to 1.59 % of the full list's height. Each item was 1.65 cm tall, corresponding to 6.3 % of the list component's height. The list was ordered and contained all numbers from 0 to 999.

A scrollbar was displayed on the rightmost edge of the list component to indicate the current position in it. Note that this information was not necessary since the list items were sequentially numbered, but we preserved it for ecological validity. On the left side of the list component, a black selection cursor, fixed vertically at the center of the list component, pointed to the item located at the vertical center of it. The pointed item was also highlighted in gray.

A trial began with the appearance of the target number on the right. To select it, participants had to scroll until the target aligns with the selection cursor and then remain stationary for 500 ms (when stationary over an item, an animation gradually changes the background color of the pointed item as feedforward of the upcoming selection). Once the target was successfully selected, the list was automatically shifted slightly to center the selection cursor precisely on the selected item. This ensured a controlled and consistent starting position for the next trial. The next target number then appeared, marking the beginning of the following trial.

The experimental task involved scrolling in two different Directions (*Up* and *Down*) to select targets located at three distinct Distances from the starting element: *Close*, which corresponded to 20% of the list component (5.24 cm) (meaning the target was already contained in the list component, and therefore visible, at the beginning of the trial), *Intermediate*, which corresponded to 200% of the list component (52.4 cm), and *Far*, which corresponded to 800% of the list component (209.6 cm).

A Block of a selection task consisted of 6 trials (2 Directions × 3 Distances). Each participants performed 6 Blocks for each Technique for a total of 144 trials per participants.

4.2.2 Counting task. The *Counting Task* required participants to count, "as quickly and accurately as possible", the number of occurrences of cat photos within a gallery of cat and dog photos, without knowing the approximate number or location of these occurrences. This design encouraged a slower, content-aware scrolling behavior that demanded cognitive effort to inspect each element. In this respect, *Counting Task* contrasts with the faster and more ballistic scrolling behavior encouraged by *Reciprocal Selection Task*. The experimental interface (illustrated Figure 6-B) displayed a gallery, which consisted of a gallery of photos of 33 rows and 3 columns. A scrollbar was displayed on the rightmost edge of the gallery component to indicate the current position in the gallery. As a reminder, the target animal, cat, was displayed as plain text on the right. The component displaying the gallery had a height and width of 21.8 cm, and each photo was displayed in a 7.3 cm square. The component therefore displayed up to 9 photos completely at once.

A trial began with the appearance of a gallery of cat and dog photos in the center of the interface. The starting position was set to the top of the gallery. The participants then scrolled through the gallery to count the number of photos of cats, and was free to scroll up or down at any time. When the participants reached the bottom of the gallery, or decided they had finished counting the photos, they pressed the space bar to display a pop-up text field and type their response using the numeric keypad. Once an answer was entered, it moved to the next trial, regardless of whether the answer was correct or not.

This task was repeated 6 times, using another of the set generated prior to the experiment, in an order counterbalanced across participants. Each of these sets was composed of 99 photos: a random number between 6 to 14 of photos of cats, and the rest of photos of dogs [3].

In total, each participants performed 6 trials of *Counting Task* for each Technique, for a total of 24 trials per participants.

4.3 Likert Questionnaire

After each task, participants were asked to answer to the following statements, presented as 7 points Likert-like items (Strongly disagree, Disagree, Somewhat disagree, Neither disagree nor agree, Somewhat agree, Agree, Strongly agree):

- The scrolling technique allowed precise control (*Precision*)
- The scrolling technique enabled efficient use (*Efficiency*)
- The scrolling technique was comfortable to use (*Comfort*)
- The scrolling technique did not cause frustration (*Frustration*)
- Overall, the scrolling technique was well suited to the task (*Suitability*)

4.4 Apparatus

We implemented the scrolling techniques and the experiment using Unity 2022.3.32f1 and a Windows 11 operating system laptop. The experiment was displayed on a Philips 27B1U5601 screen, with a diagonal length of 68.5 cm for a 2560 × 1440 pixels resolution refreshed at 75 Hz. To recognize the microgestures, we used an Optitrack system with 6 prime 13 camera placed around the interaction area (Figure 7-a). The index finger was equipped with four tracking markers (Figure 7-b) to reconstruct a virtual finger in the application. A constellation of markers was also placed on the thumb (Figure 7-b) to determine the point at which it came into contact with the index finger. The tracking coordinates were sent from the Motive 1.10.2 software to the Unity application where all contact and position calculations were performed.

To get the thumb's position on the index finger, we first estimate the thumb's projection onto the index finger. We then compute the distance between the thumb and its projection to determine if the thumb is contacting the index.

Estimating the thumb's projection onto the index finger. We model the index finger as a cubic Bezier curve. The curve has 4 control points (P_{Tip}, P_{Joint1}, P_{Joint2} and P_{Base}, Figure 7-b), and is parametrized from $t = 0$ (finger tip) to $t = 1$ (finger base) using Equation 5. These control points (corresponding to the tip, joints and base of the index finger) are computed using the 4 Optitrack markers (M_0, M_1, M_2 and M_3, Figure 7-b) placed carefully on the index so that: $P_{\text{Tip}} = M_0 + \overrightarrow{M_1 M_0}$, $P_{\text{Joint1}} = M_1$, $P_{\text{Joint2}} = M_2 + \overrightarrow{M_1 M_2}$ and $P_{\text{Base}} = M_3 + \overrightarrow{P_{\text{Joint2}} M_3}$. We approximate the thumb's projection

[3]The sets were initially randomly created using images from the Oxford-IIIT Pet Dataset. To ensure reliable distinction between the cat and dog images in these sets, 4 people were asked to classify the photos as 'cat', 'dog' or 'unclear'. All the photos misclassified or classified as unclear were discarded.

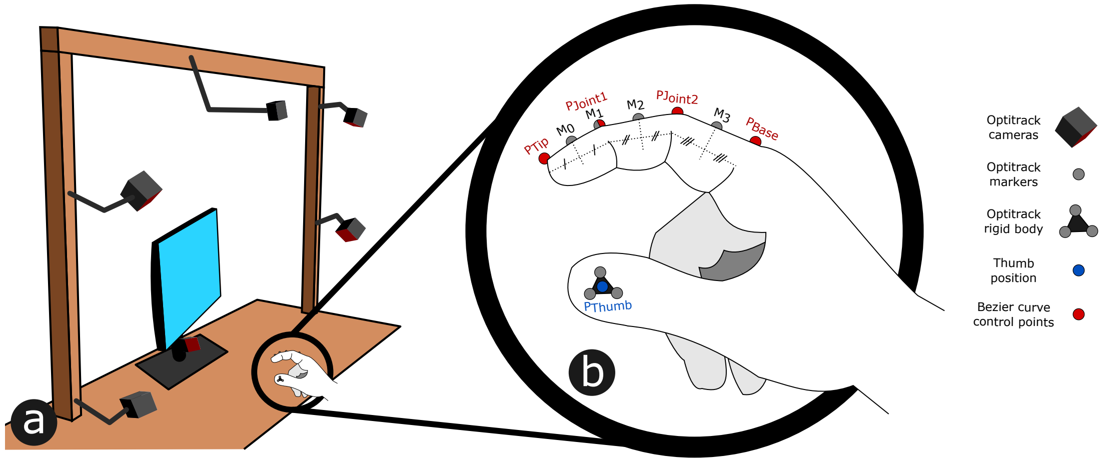

Figure 7: a) Experimental setup showing the positioning of the Optitrack cameras. b) Zoom on a participant's hand showing the positioning of the Optitrack markers and rigid body, as well as the computed control points of the cubic Bezier used for contact detection.

onto the index finger as the closest point of the curve ($P(t_{\text{Thumb}})$) to the 3D thumb position (P_{Thumb}). Thus, estimating the thumb's projection equates to estimating the parameter t_{Thumb} (i.e., the Bezier curve parameter t that minimizes the 3D distance between P_{Thumb} and $P(t)$). We use an iterative interval-refinement process. The function used to estimate the t_{Thumb} is detailed in Algorithm 1. We used these computed t_{Thumb} parameters to feed the interaction loops, as they provide a normalized position of the thumb along the index finger axis.

$$P(t) = P_{\text{Tip}} \cdot t^3 + 3 \cdot P_{\text{Joint1}} \cdot t^2 \cdot (1-t) + 3 \cdot P_{\text{Joint2}} \cdot t \cdot (1-t)^2 + P_{\text{Base}} \cdot (1-t)^3 \quad (5)$$

Contact detection. To accurately detect contacts between the thumb and the index, we ran a calibration phase with each participant, during which they dragged their thumb back and forth along the index finger. At each frame, we recorded the distance between the thumb position (P_{Thumb}) and its projection on the index ($P(t_{\text{Thumb}})$). We generated a map of 100 evenly spaced t parameters between 0 and 1, to which we associated the distance of the closest recorded t_{Thumb} parameter. This map was checked for outliers : if a distance was more than 0.05 away from its neighboring distances (in unity units), it was replaced by the average of the neighboring distances. Finally, all distances were multiplied by a 1.1 factor to provide a tolerance margin for contact detection (e.g., varying finger postures, skin elasticity, etc). During the experiment, we used this map of distances as thresholds to decide whether the thumb is in contact with the index finger or not.

Our position-detection method supports users with different finger lengths, while the calibration step ensures consistent and responsive contact detection across varying finger thicknesses.

4.5 Participants

We recruited 24 participants at the local university. Their age ranged from 20 to 40 years old (mean=24.87, std=4.94). 14 of them identify themselves as men, 10 as women. 4 self-reported as left-handed and 20 as right-handed.

4.6 Design

Our experiment used a within-subjects design with TECHNIQUES (Tap$_{Position}$, Hold$_{Rate}$, Drag$_{Position}$, Drag$_{Rate}$) as factor. Order of presentation of technique was counterbalanced across participants using balanced Latin square. Participants performed both tasks (*Reciprocal Selection Task*, *Counting Task*) with each TECHNIQUE.

For the *Reciprocal Selection Task* we collected 4 TECHNIQUES × 6 BLOCKS×3 DISTANCES×2 DIRECTIONS = 144 trials per participants. For the *Counting Task* we collected 4 TECHNIQUES × 6 TRIAL = 24 trials per participants. Each participants therefore performed 144 + 24 = 168 trials. Completing the experiment took 60 minutes on average.

4.7 Data analysis

4.7.1 Dependant variables. We collected 3 quantitative measures for the *Reciprocal Selection Task*[4]. *Completion Time* refers to the time between participants's first input and trial completion time. The *number of overshoots* refers to the number of times the black selection cursor scrolled over target. The *overshoot distance* refers to the total scrolling distance from the first overshoot for each trial (thus 0 if no overshoot).

We collected 2 quantitative measures for the *Counting Task*. *Completion Time* refers to the time taken to count all the occurrences of the target animal and type the response. *Error Count* refers to the difference between the number of cats typed by the participants and the actual number in the gallery, for each trial.

For both tasks, we collected the same subjective measures listed in section 4.3.

[4]Data for both tasks is available on https://osf.io/cpzrf/overview

```
Function estimateThumbProjection(P_Thumb):
    Function intervalRefinement(P_Thumb, t_min, t_max, stepsLeft):
        d_min = dist(P_Thumb, P(t_min));
        d_max = dist(P_Thumb, P(t_max));
        ratio = d_min/(d_min + d_max);
        t_Thumb = t_min + ratio · (t_max − t_min) ; // Refinement of t_Thumb by weighting distances to the interval bounds.
        if stepsLeft = 0 then
            return t_Thumb;
        end
        else
            new_t_min = t_Thumb − (t_max − t_min)/4 ;        // Refined lower interval bound
            new_t_max = t_Thumb + (t_max − t_min)/4 ;        // Refined upper interval bound
            if new_t_min < 0 then
                new_t_min = 0 ;                              // Bounds interval within the [0,1] range.
                new_t_max = (t_max − t_min)/2;
            end
            else if new_t_max > 1 then
                new_t_min = 1 − (t_max − t_min)/2;
                new_t_max = 1;
            end
            return intervalRefinement(P_Thumb, new_t_min, new_t_max, stepsLeft − 1);
        end
    end
    return intervalRefinement(P_Thumb, 0, 1, 10);
end
```

Algorithm 1: Algorithm estimating the thumb's projection Bezier curve parameter onto the index finger

4.7.2 Analysis. Following prior recommendations [41], and given that we have fewer than 25 participants, we used the geometric mean to estimate the population center for *Completion Time*.

Quantitative measures are analyzed using repeated-measure ANOVAs conducted with the `aov_ez` function from the `afex` R package in R Studio. We report Greenhouse-Geisser corrected degrees of freedom and p-values when sphericity was violated. When a significant effect was observed, we conducted post-hoc pairwise comparisons using the `emmeans` package.

With regard to subjective measures, we analyzed each question using a Friedman test to detect any significant effects. If any were identified, we conducted post-hoc pairwise comparisons using Wilcoxon tests with Bonferroni correction.

Unless specified otherwise, errors bars on charts corresponds to bootstrapped 95 % confidence intervals (CIs). We report results using the mean (or geometric mean) along with bootstrapped 95 % confidence intervals, formatted as $m = mean, CI = [lowerCI, upperCI]$.

5 Results

5.1 Reciprocal Selection task

5.1.1 Preliminary analysis. Our preliminary analysis, a repeated-measures ANOVA on completion time, revealed a significant effect of Block ($F_{3.61,83.14} = 9.44$, $p = 0.001$, $\eta^2_G = 0.17$). The post-hoc pairwise comparisons revealed a significant difference between Block 1 and all the others Blocks. As our focus is on assessing stable, trained performance, we exclude the first block from further analysis. All results reported below are therefore based on the remaining blocks.

5.1.2 Target selection time. Overall, selecting targets (see Figure 8-a) using Hold$_{Rate}$ took significantly less time than with Tap$_{Position}$. Indeed, a repeated-measures ANOVA revealed significant effects of all factors. For Technique ($F_{2.08,47.90} = 3.87$, $p = 0.05$, $\eta^2_G = 0.05$), post-hoc comparisons revealed a significant difference between Tap$_{Position}$ ($m = 6.78, CI = [6.24, 7.69]$) and Hold$_{Rate}$ ($m = 5.59, CI = [5.16, 6.1]$), and no other. For Direction ($F_{1,23} = 8.82$, $p = 0.01$, $\eta^2_G = 0.01$), a significant difference was found between Up ($m = 7.03, CI = [6.46, 7.59]$) and Down ($m = 6.56, CI = [6.09, 7.03]$). For Distance ($F_{1.51,34.84} = 265.24$, $p = 0.001$, $\eta^2_G = 0.48$), significant differences between all distances were found: *Close* ($m = 4.25, CI = [3.93, 4.57]$), *Intermediate* ($m = 6.45, CI = [5.93, 6.97]$) and *Far* ($m = 9.68, CI = [8.91, 10.46]$).

5.1.3 Number of Overshoot. Hold$_{Rate}$ resulted in significantly less overshoots, while Drag$_{Position}$ resulted in significantly more (see Figure 9). The Distances also have an effect, small target distance (*Close*) produce significantly more overshoot than longer distance (*Intermediate* and *Far*). However this Distance effect only impact Hold$_{Rate}$, Drag$_{Position}$ and Drag$_{Rate}$ techniques, Tap$_{Position}$ number of overshoot is not affected by Distance. Indeed, a repeated-measures ANOVA revealed significant effects of Technique, Distance, and their interaction. For Technique ($F_{2.58,59.32} = 33.87$, $p = 0.001$, $\eta^2_G = 0.318$), post-hoc comparisons revealed that Tap$_{Position}$ ($m = 1.32, CI = [1.11, 1.67]$) is significantly different than Hold$_{Rate}$ ($m = 0.56, CI = [0.5, 0.64]$) (p<0.001) and Drag$_{Position}$ ($m = 1.91, CI = [1.71, 2.14]$) (p<0.01), and that Hold$_{Rate}$ is significantly different than Drag$_{Position}$ (p<0.001) and Drag$_{Rate}$ ($m = 1.07, CI = [0.88, 1.28]$) (p<0.001) and that Drag$_{Position}$

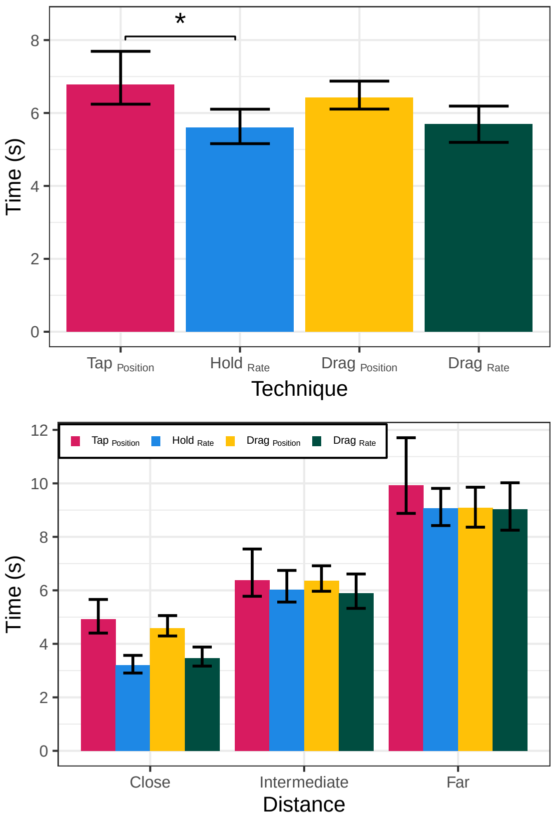

Figure 8: Reciprocal Selection Task: geometric mean completion times overall (top) and per distance (bottom), with bootstrapped 95 % **CIs.**

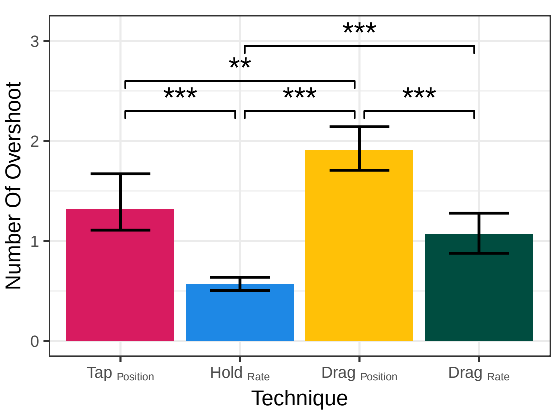

Figure 9: Reciprocal Selection Task: mean of number of overshoot with bootstrapped 95 % **CIs.**

is significantly different than Drag$_{Rate}$ (p<0.001). For Distance ($F_{1.86,42.71} = 16.54$, $p = 0.001$, $\eta^2_G = 0.055$), post-hoc pairwise analysis revealed that *Close* ($m = 0.99, CI = [0.85, 1.1]$) is significantly different than *Intermediate* ($m = 1.27, CI = [1.11, 1.44]$) (p<0.01) and *Far* ($m = 1.39, CI = [1.24, 1.51]$) (p<0.001). For Technique × Distance interaction effect ($F_{4.40,101.27} = 3.85$, $p = 0.01$, $\eta^2_G = 0.028$), post-hoc analysis revealed almost the same significant differences between techniques, although with varying effect sizes. The exceptions are for *Close* Distance: there was no significant difference between Tap$_{Position}$ and Drag$_{Position}$, while a significant difference was found between Tap$_{Position}$ and Drag$_{Rate}$ (p<0.001).

5.1.4 *Overshoot distance.*

Techniques did not result in significantly different overshoot distances, but generally, a *Far* distance target implies a greater overshoot distance than a *Close* or a *Intermediate* distance target, Indeed, the repeated-measures ANOVA revealed a significant effect of Distance ($F_{1.26,28.92} = 6.64$, $p = 0.05$, $\eta^2_G = 0.004$). The post-hoc pairwise comparisons showed that *Far* ($m = 8.66cm, CI = [7.19, 10.81]$) is significantly different than *Close* ($m = 4.79cm, CI = [3.83, 6.65]$) (p<0.05) and *Intermediate* ($m = 6.09cm, CI = [5.21, 8.03]$) (p<0.001).

5.1.5 *Subjective Assessment.*

According to the subjective assessments (see Figure 10), Hold$_{Rate}$ and Drag$_{Rate}$ were preferred by participants. Conversely, Tap$_{Position}$ was the least preferred.

Participants felt having a more precise control (see Figure 10) with Hold$_{Rate}$ and Drag$_{Rate}$ than with Tap$_{Position}$ and Drag$_{Position}$. Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=24.956, $p < 0.01$). The post-hoc test showed that Tap$_{Position}$ is significantly different than Hold$_{Rate}$ (p<0.05, r=0.42) and Drag$_{Rate}$ (p<0.01, r=0.50) and that Drag$_{Position}$ is significantly different than Hold$_{Rate}$ (p<0.05, r=0.43) and Drag$_{Rate}$ (p<0.01, r=0.48).

Participants reported feeling more efficient (see Figure 10) with Hold$_{Rate}$ and than with Tap$_{Position}$ and with Drag$_{Rate}$ than with Tap$_{Position}$ and Drag$_{Position}$. Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=24.956, $p < 0.01$). The post-hoc test showed that Tap$_{Position}$ is significantly different than Hold$_{Rate}$ (p<0.01, r=0.45) and Drag$_{Rate}$ (p<0.01, r=0.54) and that Drag$_{Position}$ is significantly different than Drag$_{Rate}$ (p<0.05, r=0.43).

Hold$_{Rate}$ was reported as more comfortable to use than Tap$_{Position}$ and Drag$_{Position}$, and Drag$_{Rate}$ more thant Tap$_{Position}$ (see Figure 10). Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=25.393, $p < 0.01$). The post-hoc test showed that Tap$_{Position}$ is significantly different than Hold$_{Rate}$ (p<0.01, r=0.50) and Drag$_{Rate}$ (p<0.05, r=0.45) and that Hold$_{Rate}$ is significantly different than Drag$_{Position}$ (p<0.05, r=0.44).

Participants reported feeling less frustrated (see Figure 10) with Hold$_{Rate}$ and Drag$_{Rate}$ than with Tap$_{Position}$ and Drag$_{Position}$. Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=18.206, $p < 0.01$). The post-hoc test showed that Tap$_{Position}$ is significantly different than Hold$_{Rate}$ (p<0.05, r=0.42) and Drag$_{Rate}$ (p<0.05, r=0.43) and that Drag$_{Position}$ is significantly different than Hold$_{Rate}$ (p<0.05, r=0.44) and Drag$_{Rate}$ (p<0.05, r=0.40).

Finally, participants perceived Hold$_{Rate}$ and Drag$_{Rate}$ more suited to a *Reciprocal Selection Task* than Tap$_{Position}$ (see Figure 10). Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=17.653, $p < 0.01$). The post-hoc test showed that Tap$_{Position}$ is significantly different than Hold$_{Rate}$ (p<0.05, r=0.40) and Drag$_{Rate}$ (p<0.01, r=0.50).

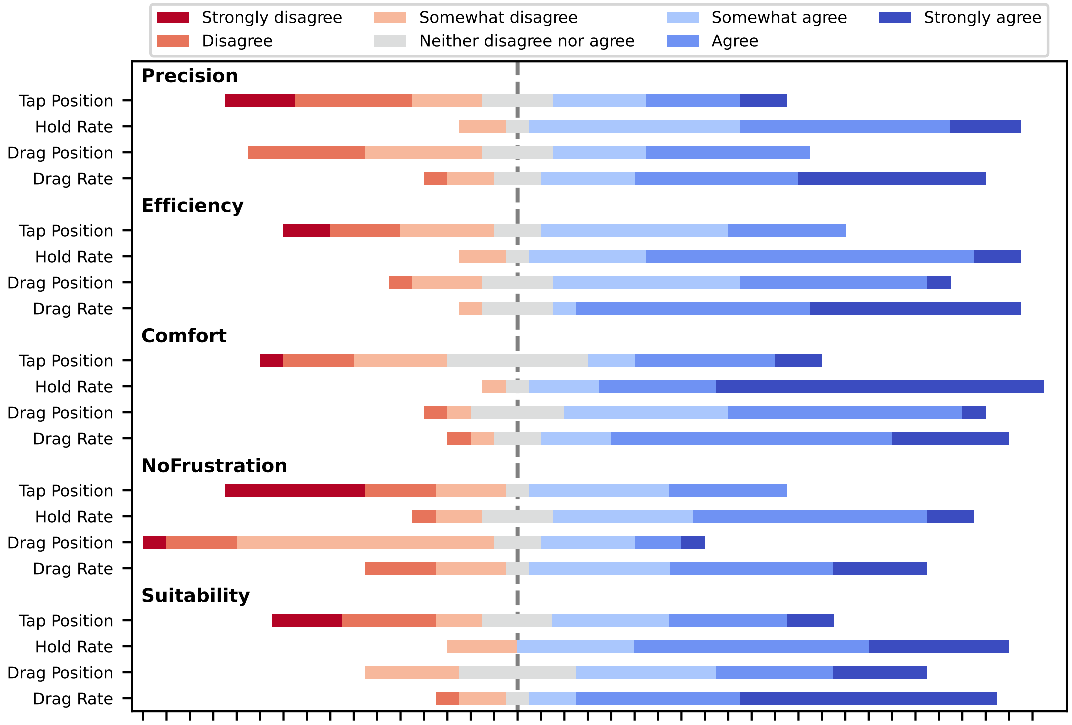

Figure 10: Reciprocal Selection Task: Results on 7 points Likert scale for Precision, Efficiency, Comfortability, NoFrustration and Suitability.

5.2 Counting task

5.2.1 Time. Completing *Counting Task* (see Figure 11) took less time with Hold_{Rate} and Drag_{Rate} than with $\text{Tap}_{Position}$ and $\text{Drag}_{Position}$. More precisely, a repeated-measures ANOVA revealed significant effects of Technique. For Technique ($F_{1.96,44.97} = 7.73$, $p = 0.001$, $\eta^2_G = 0.056$), post-hoc pairwise analysis revealed that $\text{Tap}_{Position}$ ($m = 44.28, CI = [40.16, 47.99]$) is significantly different than Hold_{Rate} ($m = 37.9, CI = [34.65, 41.29]$) (p<0.01) and Drag_{Rate} ($m = 36.25, CI = [32.73, 40.73]$) (p<0.001) and that Hold_{Rate} is significantly different than $\text{Drag}_{Position}$ ($m = 44.6, CI = [39.2, 49.84]$) (p<0.05) and that $\text{Drag}_{Position}$ is significantly different than Drag_{Rate} (p<0.05).

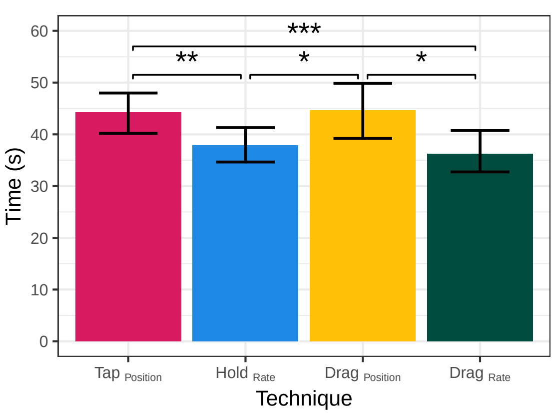

Figure 11: Counting Task: geometric mean completion times with bootstrapped 95 % CIs.

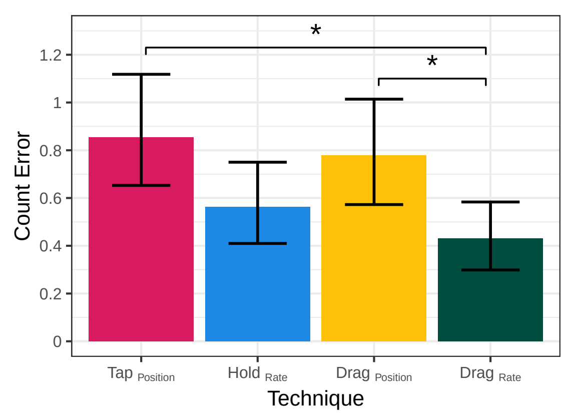

Figure 12: Counting Task: mean count error with bootstrapped 95 % CIs.

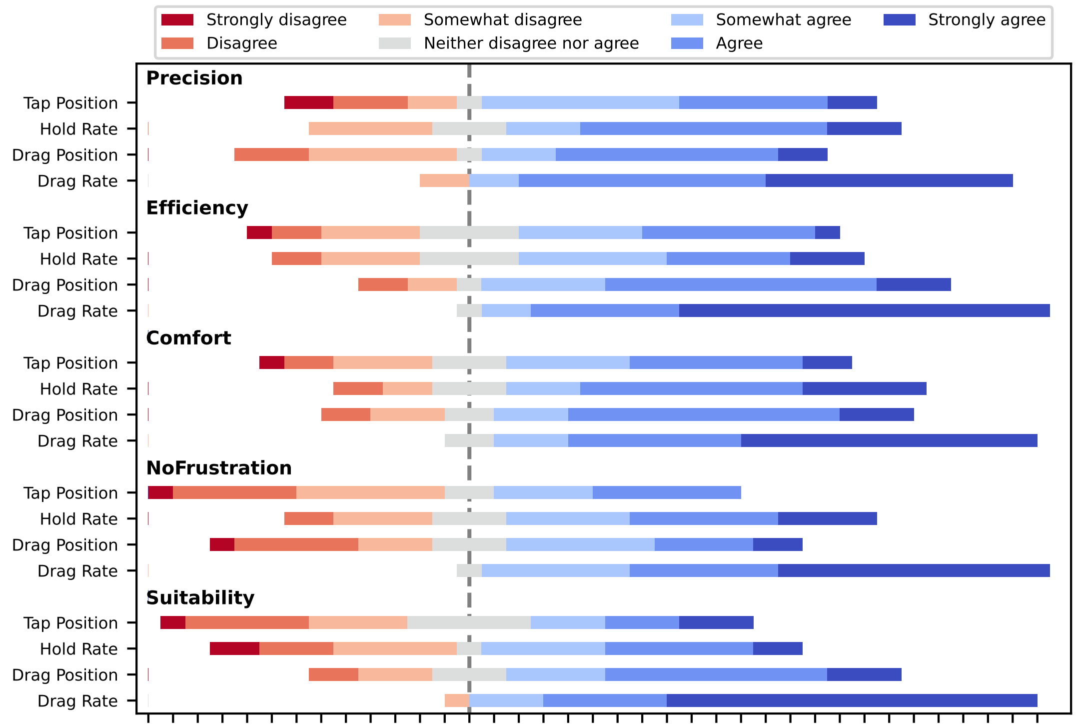

Figure 13: Counting Task: Results on 7 points Likert scale for Precision, Efficiency, Comfortability, NoFrustration and Suitability.

5.2.2 Counts Error. When counting cats (see Figure 12), Drag_{Rate} results in less error than $\text{Tap}_{Position}$ and $\text{Drag}_{Position}$. More precisely, a repeated-measures ANOVA revealed significant effects for Technique factor ($F_{2.31,53.10} = 5.65$, $p = 0.005$, $\eta^2_G = 0.104$). The post-hoc pairwise comparisons showed that Drag_{Rate} ($m = 0.43, CI = [0.3, 0.58]$) is significantly different than $\text{Tap}_{Position}$ (m=0.85, CI= $[0.65, 1.12]$) (p<0.05) and $\text{Drag}_{Position}$ ($m = 0.78, CI = [0.57, 1.01]$) (p<0.05).

5.2.3 Subjective assessment. According to the subjective assessments (see Figure 13), Drag_{Rate} are significantly more appreciated than the three others among participants.

Participants perceived a more precise control (Figure 13) with Drag_{Rate} than with $\text{Tap}_{Position}$ and $\text{Drag}_{Position}$. Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=17.694, $p < 0.01$). The post-hoc test showed that Drag_{Rate} is significantly different than $\text{Tap}_{Position}$ (p<0.05, r=0.46) and $\text{Drag}_{Position}$ (p<0.05, r=0.43).

Participants reported feeling more efficient (Figure 13) with Drag_{Rate} than with the other techniques. Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=25.479, $p < 0.01$). The post-hoc test showed that Drag_{Rate} is significantly different than $\text{Tap}_{Position}$ (p<0.01, r=0.56), Hold_{Rate} (p<0.01, r=0.48) and $\text{Drag}_{Position}$ (p<0.05, r=0.46).

Participants also reported perceiving Drag_{Rate} more comfortable than $\text{Tap}_{Position}$ (Figure 13). Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=19.613, $p < 0.01$). The post-hoc test showed significant differences between $\text{Tap}_{Position}$ and Drag_{Rate} (p<0.01, r=0.52).

Participants also reported feeling less frustrated (Figure 13) with Drag_{Rate} than with the other techniques. Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=23.113, $p < 0.01$). The post-hoc test showed that Drag_{Rate} is significantly different than $\text{Tap}_{Position}$ (p<0.01, r=0.57), Hold_{Rate} (p<0.05, r=0.40) and $\text{Drag}_{Position}$ (p<0.01, r=0.52).

Finally, participants reported finding h Drag_{Rate} more suited to the counting task than all other techniques (Figure 13). Indeed, a Friedman test revealed a significant effect of Technique ($X^2(3)$=30.806, $p < 0.01$). The post-hoc test showed that Drag_{Rate} is significantly different than $\text{Tap}_{Position}$ (p<0.01, r=0.52), Hold_{Rate} (p<0.01, r=0.53) and $\text{Drag}_{Position}$ (p<0.01, r=0.49).

5.3 Results Summary

Results are summarized table 14.

Reciprocal Selection task. In the *Reciprocal Selection Task*, Hold_{Rate} required less time to complete the trials than $\text{Tap}_{Position}$. Hold_{Rate} resulted in fewer overshoots than all other techniques, whereas $\text{Drag}_{Position}$ resulted in more. There were two exceptions for close distances: no significant difference was found between $\text{Tap}_{Position}$ and $\text{Drag}_{Position}$, and a significant difference was found between $\text{Tap}_{Position}$ and Drag_{Rate}. However, the techniques had no

Task	Metric	Tap$_{Position}$	Hold$_{Rate}$	Drag$_{Posistion}$	Drag$_{Rate}$
Reciprocal Selection	Time (in s)	m = 6.78, CI = [6.24, 7.69]	m = 5.59, CI = [5.16, 6.10]	m = 6.42, CI = [6.11, 6.87]	m = 5.69, CI = [5.19, 6.19]
	#Overshoots	m = 1.32, CI = [1.11, 1.67]	m = 0.56, CI = [0.50, 0.64]	m = 1.91, CI = [1.71, 2.14]	m = 1.07, CI = [0.88, 1.28]
	Overshoot distance	m = 6.68, CI = [5.05, 10.51]	m = 5.40, CI = [4.10, 9.32]	m = 6.58, CI = [5.57, 8.21]	m = 5.96, CI = [4.35, 8.81]
	Precision	neither disagree or agree	agree	neither disagree or agree	agree
	Efficiency	somewhat agree	agree	somewhat agree	agree
	Comfort	neither disagree or agree	strongly agree	somewhat agree	agree
	No Frustration	neither disagree or agree	agree	somewhat disagree	somewhat agree
	Suitability	somewhat agree	agree	somewhat agree	agree
Counting	Time (in s)	m = 44.28, CI = [40.16, 47.99]	m = 37.90, CI = [34.65, 41.29]	m = 44.6, CI = [39.20, 49.84]	m = 36.25, CI = [32.73, 40.73]
	#Errors	m = 0.85, CI = [0.65, 1.12]	m = 0.56, CI = [0.41, 0.75]	m = 0.78, CI = [0.57, 1.01]	m = 0.43, CI = [0.30, 0.58]
	Precision	somewhat agree	agree	somewhat agree	agree
	Efficiency	somewhat agree	somewhat agree	agree	strongly agree
	Comfort	somewhat agree	agree	agree	strongly agree
	No Frustration	neither disagree or agree	somewhat agree	somewhat agree	agree
	Suitability	neither disagree or agree	somewhat agree	agree	strongly agree

Legend

for quantitative data

m = mean (geometrical mean for Time)

CI = [lower 95% CI, upper 95% CI]

for qualitative data

cell colour = median value (strongly disagree , disagree , somewhat disagree , neither disagree or agree , somewhat agree , agree , strongly agree)

● = show significant difference with other techniques, each ● represents a technique (see columns' name).
For instance, cell shows significant differences with **Hold$_{Rate}$** and **Drag$_{Posistion}$**)

Figure 14: Results summary of all quantitative (geometric mean and 95% CIs) and qualitative (median Likert value) data for both experiment. Significative differences are reported using colored corners.

significant effect on overshoot distances. Regarding subjective assessments, participants reported having a more precise control and felt less frustrated with Hold$_{Rate}$ and Drag$_{Rate}$ than with Tap$_{Position}$ and Drag$_{Position}$. They also felt more efficient with Hold$_{Rate}$ than with Tap$_{Position}$, and with Drag$_{Rate}$ more than with both Tap$_{Position}$ and Drag$_{Position}$. Hold$_{Rate}$ was rated as more comfortable to use than Tap$_{Position}$ and Drag$_{Position}$, and Drag$_{Rate}$ as more comfortable than Tap$_{Position}$. Finally, participants perceived Hold$_{Rate}$ and Drag$_{Rate}$ as more suitable for the *Reciprocal Selection Task* than Tap$_{Position}$.

Counting task. Completing the *Counting Task* took less time with Hold$_{Rate}$ and Drag$_{Rate}$ than with Tap$_{Position}$ and Drag$_{Position}$. Anecdotally, Drag$_{Rate}$ resulted in fewer errors than Tap$_{Position}$ and Drag$_{Position}$ when counting cats. Regarding subjective assessments, participants perceived Drag$_{Rate}$ as offering more precise control than Tap$_{Position}$ and Drag$_{Position}$. They also rated Drag$_{Rate}$ as more comfortable than Tap$_{Position}$. Participants reported feeling more efficient and less frustrated with Drag$_{Rate}$ than with the other techniques, and considered it the most suitable for the task.

6 Discussion

We designed 4 techniques that support scrolling using microgestures, and contrasted their performance in a study involving 2 different scrolling tasks to obtain more ecological results. The first task, *Reciprocal Selection Task*, simulated situations where users knew the content to scroll through and the stopping point, requiring them to scroll as quickly as possible to a target. The second task, *Counting Task*, simulated a browsing/search scenario, where users had to move slowly to enable content-aware scrolling. With these tasks, we aimed to evaluate and compare performance, effectiveness, and usability in different contexts.

6.1 Scrolling

These two tasks represent distinct behaviors that frequently occur while scrolling through content in everyday life. We assumed that the microgesture **Input Type** would influence participants' preferences and performance. In particular, we hypothesized that a fully controlled continuous **Input Type** technique would produce better results than a partially controlled continuous **Input Type** or a discrete **Input Type** (**H1** & **H2**).

The results encourage us to partially reject **H1**. Although we hypothesized that techniques based on Drag would outperform the others, this was not systematically the case: Drag$_{Rate}$ stands out as the most effective, but Drag$_{Position}$ was among the least efficient. Regarding **H2**, we hypothesized that Hold$_{Rate}$ would result in better performance than Tap$_{Position}$. In this respect, **H2** is supported. Moreover, Hold$_{Rate}$ also appears to perform better than Drag$_{Position}$. Finally, as expected, Tap$_{Position}$, using a discrete **Input Type**, performs the worst.

When looking performance on specific tasks, Hold$_{Rate}$ is most effective for reaching a specific target with partial knowledge of its location in the list (e.g., selecting an application or a song in a playlist), whereas Drag$_{Rate}$ is best suited for browsing content (e.g., scanning an image gallery or locating a section on a website). The participants appreciated being able to adjust the scrolling speed with Drag_{Rate}, making it the preferred technique for browsing, as also reflected in subjective ratings. Both techniques are strong alternatives: if only one is implemented, we recommend Drag_{Rate}, since it excels in browsing tasks and we did not find evidence that it underperforms compared to Hold_{Rate} for target selection in known lists. If continuous position tracking is difficult to implement, then Hold_{Rate} remains a solid option.

We also hypothesized that users would perform better with $\text{Drag}_{Position}$, given its similarity to the flick gestures used on phones and trackpads. However, results showed that $\text{Drag}_{Position}$ was among the least effective of the four techniques, thus rejecting **H3**. This unexpected outcome may be explained by several factors. First, although the principle of the technique resembles frequent flick gestures on smartphone, the interactive surface differs: the finger tends to stick and is a non-flat surface, hence it slides less smoothly and continuously than a flat tactile surface like a touchscreen. Second, in the case of thumb-based DRAGS and flicks (e.g., on phones), scrolling using thumb-to-index microgesture typically requires movements along a slightly different axis because of hands' biomechanical constrains, which may have affected dexterity. Third, the transfer function could also play a role; despite refinements made by the author and four pilot users, further optimization may be possible. Finally, it is also possible that the other techniques are simply better suited to this type of scrolling task.

6.2 When scrolling through familiar content

When users are familiar with the content and have an idea of their target's location, Hold_{Rate} proves to be the most efficient technique. Drag_{Rate} do not seem less efficient than Hold_{Rate} and could therefore be an acceptable alternative. In terms of selection time, the only significant difference we observed was that Hold_{Rate} outperforms $\text{Tap}_{Position}$. But regarding overshoot, Hold_{Rate} produces the fewest errors, while $\text{Drag}_{Position}$ causes significantly more overshoot than the other techniques making it the least convenient for target selection. Overshoot distance itself, however, shows no significant differences across techniques. Subjective assessments align with these results: Hold_{Rate} and Drag_{Rate} received the highest ratings, $\text{Tap}_{Position}$ was consistently rated the lowest, and $\text{Drag}_{Position}$ received mixed evaluations.

6.3 When scrolling through unfamiliar content

When users are not familiar with the content and do not know the target's location, Drag_{Rate} emerges as the most effective technique. However, if continuous position tracking is not feasible, Hold_{Rate} provides a acceptable alternative. Although Hold_{Rate} does not outperform the other techniques in terms of count errors or subjective ratings, it resulted in lower completion times. Both Hold_{Rate} and Drag_{Rate} significantly reduce browsing time compared to $\text{Tap}_{Position}$ and $\text{Drag}_{Position}$. Moreover, Drag_{Rate} also leads to fewer count errors than $\text{Tap}_{Position}$ and $\text{Drag}_{Position}$, combining efficiency with accuracy. Subjective ratings confirm these findings: participants perceived Drag_{Rate} as more precise, more efficient, less frustrating, and overall better suited to the task than the other techniques.

6.4 Limitations and Future Work

We now discuss limitations of our work, and discuss how it also opens up avenues for future work.

Transfer Function. Designing the scrolling techniques relying on microgestures required us to decide on precise transfer functions and associated parameters to map microgestures to scrolling operations. In order to avoid this decision being arbitrary and limited to the designer's initial intent, we first iterated over the parameter until satisfied overall, and then involved 4 additional pilot users to refine it further. We believe this procedure helped us to reach relatively optimal designs, notably because the adjustments required by the additional pilot users were minimal, giving us confidence in the reliability of the first refined values. We acknowledge that we did not explore every possible transfer function, and that alternatives could also be relevant. Typically, operating systems allow users to customize the sensitivity of scrolling transfer functions to their preference [1, 33, 39]. However, in order to keep our experiments within bearable durations, design choices had to be made, and we selected the functions presented in this paper. While we did the best we could, fine-tuning the transfer functions through pilot experiments as described in section 3.2.5, exploring other transfer functions remains an interesting direction for future work. While transfer functions could always be modified, we thus consider them sufficiently robust for the purposes of this exploratory study on scrolling with microgestures, which therefore gives us confidence in our results.

Limitation of vision-based microgestures sensing. This study was conducted using a custom vision-based detection system. Although neither we nor the participants noticed detection errors or high latency, the results could still slightly differ with another sensing technology. For position detection, we approximated finger position based on visual markers placed at proportional distances along the index finger. In principle, the detected position should be correctly and uniformly distributed across the finger surface. However, issues such as positional noise or sudden value jumps could affect the techniques —particularly those based on Drag microgestures. Contact detection also presents limitations. Imprecise or delayed contact detection could introduce latency when stopping rate-based techniques, or could segment drag distance incorrectly in the case of $\text{Drag}_{Position}$. But again, participants did not report any specific issue in this regard. Therefore, there is no reason to believe that results would be sigificantly differet with other vision-based tracking systems. It is also important to note that these techniques were tested only with vision-based detection, with no physical obstruction on the contact surface. Results might slightly differ if a glove was used for sensing. Indeed, a glove removes skin-to-skin contact, which normally provides natural tactile feedback, and may change how the thumb slides or sticks, potentially affecting the behavior of drag-based techniques. This difference could be an interesting aspect to investigate further.

$\text{Tap}_{Position}$ Displacement. An additional consideration concerns the discrete content displacement of $\text{Tap}_{Position}$, which differs from the continuous content displacement of the other techniques. 2 participants suggested that incorporating an animation could make $\text{Tap}_{Position}$ more comfortable visually, and considered it may also

make it more efficient. However, given the overall results, even with this modification we do not anticipate that $\text{Tap}_{Position}$ would be significantly more efficient than Hold_{Rate} or Drag_{Rate}.

Comparisons with previous scrolling techniques. This work focused on exploring and evaluating microgesture techniques as part of completing the set of interactions associated with the microgesture modality. Rather than positioning these techniques as the best overall solution for scrolling, our goal was to better understand their potential and limitations. We see this as a foundation for future work, which could extend the investigation by comparing microgesture-based scrolling with other scrolling techniques across different input modalities.

Hybrid Scrolling Technique. In addition, we are interested in exploring a hybrid technique combining Hold_{Rate} and Drag_{Rate}. These techniques should be compatible: Hold_{Rate} uses the proximal and distal phalanges as a trigger, while Drag_{Rate} uses the middle phalanx. Users could employ Hold_{Rate} for scrolling across known content and making fine adjustments, while Drag_{Rate} would serve as the primary method for exploring content.

Scrolling on another finger or on multiple fingers. An interesting direction for future work concerns the choice of fingers used for microgestures. A core principle of microgestures is that different fingers can serve as different input channels, but scrolling is unlikely to be the primary interaction in an interaction mapping. In this context, it may be preferable not to rely on the index finger for scrolling. Prior work found that thumb contact comfort is slightly lower on the middle finger than on the index finger, and even lower on the ring finger [18]. Although we did not study scrolling techniques with different fingers, prior work has shown that microgestures can be transferred across fingers such as the index, middle, ring, and little finger [24]. In our case, the interacting finger would still be the thumb. Under this assumption, we expect performance to be comparable with the middle finger and less efficient with the ring finger. As transferability also depends on the interaction itself [24], we do not expect good scrolling performance on other fingers that lack the thumb's dexterity. Further investigation is needed to confirm these expectations. A second interesting direction for future work is the use of multiple fingers for scrolling. A scrolling technique could be enhanced by assigning different mappings to different fingers. For example, one finger could be used for absolute scrolling or to scroll through the chapters of a document, whereas a second finger would be dedicated to precise scrolling. Alternatively, different fingers could be used with varying scrolling transfer functions or precision levels. A third direction for future work is to investigate the potential of using multiple fingers for 2D scrolling. Possible solutions include assigning different fingers to different axes, or joining several fingers to create a surface on which the thumb could move in 2D.

Microgestures as a fully independent interaction modality. As we already mentioned, the goal of this study was to enrich a microgesture based input vocabulary with scrolling interaction capabilities. Future work should now investigate the creation of a comprehensive microgesture mapping that supports a wide range of interactions. A first step in this direction will be to clearly identify and define the basic interactions needed, and classify them according to their importance. As discussed previously, multiple fingers can be used for microgestures, and the assignment of fingers to actions should depend on the importance of the action. Most of the interactions studied so far are discrete and concern menu navigation or command selection. The index finger could then be dedicated to these: tip phalanx for "previous item" central phalanx for "select" and proximal phalanx for "next item". To support continuous interactions, the middle finger could be used for scrolling and/or slider manipulation. The ring finger could then provide a second continuous area for 2D scrolling or pointing when combined with the middle finger. If more discrete interactions are needed, they could be assigned to the little finger. Although dexterity is more limited on little finger, discrete interactions on the little finger are more practical than continuous interactions. All of the above represents only an initial concept to be explored and will be the focus of our future work. Finally, user interfaces should continue to consider how these input possibilities can be presented to users to support their discovery and learning [30, 31].

7 Conclusion

In this paper, we designed and evaluated 4 microgesture-based scrolling techniques. Microgestures represent a promising interaction modality, as they are subtle movements that can be performed without involving the arm or wrist. Despite these advantages, prior research has focused almost exclusively on discrete interactions. Through a user study with 24 participants, we demonstrated the viability of microgestures as a continuous input modality, and in particular their reliability for scrolling. Taken together, our results indicate that microgestures can effectively support both target selection in known lists and content-aware browsing, particularly with the Drag_{Rate} technique, which involves dragging the thumb along the index finger to control scrolling speed. This work constitutes a significant step toward establishing microgestures as a fully independent interaction modality.

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References

[1] J. Aceituno, S. Malacria, P. Quinn, N. Roussel, A. Cockburn, and G. Casiez. 2017. The design, use, and performance of edge-scrolling techniques. *International Journal of Human-Computer Studies* 97 (2017), 58–76. https://doi.org/0.1016/j.ijhcs.2016.08.001

[2] Dzimitry Aliakseyeu, Pourang Irani, Andrés Lucero, and Sriram Subramanian. 2008. Multi-flick: an evaluation of flick-based scrolling techniques for pen interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) *(CHI '08)*. Association for Computing Machinery, New York, NY, USA, 1689–1698. https://doi.org/10.1145/1357054.1357319

[3] Axel Antoine, Sylvain Malacria, and Géry Casiez. 2017. ForceEdge: Controlling Autoscroll on Both Desktop and Mobile Computers Using the Force. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) *(CHI '17)*. Association for Computing Machinery, New York, NY, USA, 3281–3292. https://doi.org/10.1145/3025453.3025605

[4] Caroline Appert and Jean-Daniel Fekete. 2006. OrthoZoom scroller: 1D multi-scale navigation. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Montréal, Québec, Canada) *(CHI '06)*. Association for Computing Machinery, New York, NY, USA, 21–30. https://doi.org/10.1145/1124772.1124776

[5] Mathias Baglioni, Sylvain Malacria, Eric Lecolinet, and Yves Guiard. 2011. Flick-and-brake: finger control over inertial/sustained scroll motion. In *CHI '11 Extended Abstracts on Human Factors in Computing Systems* (Vancouver, BC, Canada) *(CHI EA '11)*. Association for Computing Machinery, New York, NY, USA, 2281–2286. https://doi.org/10.1145/1979742.1979853

[6] Charles Bailly and Julien Castet. 2023. WIM-Drone: Combining a World-In-Miniature and Micro-Gestures for Teleoperation in Augmented Reality. In *Proceedings of the 34th Conference on l'Interaction Humain-Machine* (TROYES, France) *(IHM '23)*. Association for Computing Machinery, New York, NY, USA, Article 11, 11 pages. https://doi.org/10.1145/3583961.3583972

[7] Anne Katrine Blangsted, Gisela Sjøgaard, Pascal Madeleine, Henrik Baare Olsen, and Karen Søgaard. 2005. Voluntary low-force contraction elicits prolonged low-frequency fatigue and changes in surface electromyography and mechanomyography. *Journal of Electromyography and Kinesiology* 15, 2 (2005), 138–148. https://doi.org/10.1016/j.jelekin.2004.10.004

[8] Roger Boldu, Alexandru Dancu, Denys J.C. Matthies, Pablo Gallego Cascón, Shanaka Ransir, and Suranga Nanayakkara. 2018. Thumb-In-Motion: Evaluating Thumb-to-Ring Microgestures for Athletic Activity. In *SUI '18: Proceedings of the 2018 ACM Symposium on Spatial User Interaction* (Berlin, Germany) *(SUI '18)*. Association for Computing Machinery, New York, NY, USA, 150–157. https://doi.org/10.1145/3267782.3267796

[9] Adrien Chaffangeon Caillet, Alix Goguey, and Laurence Nigay. 2025. Microgesture Grasp: A journey from human capabilities to interaction with microgestures. *International Journal of Human-Computer Studies* 195 (2025), 103398. https://doi.org/10.1016/j.ijhcs.2024.103398

[10] Géry Casiez and Daniel Vogel. 2008. The effect of spring stiffness and control gain with an elastic rate control pointing device. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Florence, Italy) *(CHI '08)*. Association for Computing Machinery, New York, NY, USA, 1709–1718. https://doi.org/10.1145/1357054.1357321

[11] Géry Casiez, Daniel Vogel, Qing Pan, and Christophe Chaillou. 2007. RubberEdge: reducing clutching by combining position and rate control with elastic feedback. In *Proceedings of the 20th Annual ACM Symposium on User Interface Software and Technology* (Newport, Rhode Island, USA) *(UIST '07)*. Association for Computing Machinery, New York, NY, USA, 129–138. https://doi.org/10.1145/1294211.1294234

[12] Adrien Chaffangeon Caillet, Alix Goguey, and Laurence Nigay. [n. d.]. 3D Selection in Mixed Reality: Designing a Two-Phase Technique To Reduce Fatigue. In *IEEE ISMAR 2023*. https://doi.org/10.1109/ISMAR59233.2023.00095

[13] Adrien Chaffangeon Caillet, Alix Goguey, and Laurence Nigay. 2023. μGlyph: a Microgesture Notation. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 269, 28 pages. https://doi.org/10.1145/3544548.3580693

[14] Edwin Chan, Teddy Seyed, Wolfgang Stuerzlinger, Xing-Dong Yang, and Frank Maurer. [n. d.]. User Elicitation on Single-hand Microgestures. In *ACM CHI 2016*. https://doi.org/10.1145/2858036.2858589

[15] Andy Cockburn, Philip Quinn, Carl Gutwin, and Stephen Fitchett. 2012. Improving scrolling devices with document length dependent gain. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) *(CHI '12)*. Association for Computing Machinery, New York, NY, USA, 267–276. https://doi.org/10.1145/2207676.2207714

[16] Gauthier Robert Jean Faisandaz, Alix Goguey, Christophe Jouffrais, and Laurence Nigay. 2022. Keep in Touch: Combining Touch Interaction with Thumb-to-Finger μGestures for People with Visual Impairment. In *Proceedings of the 2022 International Conference on Multimodal Interaction* (Bengaluru, India) *(ICMI '22)*. Association for Computing Machinery, New York, NY, USA, 105–116. https://doi.org/10.1145/3536221.3556589

[17] Gauthier Robert Jean Faisandaz, Alix Goguey, Christophe Jouffrais, and Laurence Nigay. 2023. μGeT: Multimodal eyes-free text selection technique combining touch interaction and microgestures. In *Proceedings of the 25th International Conference on Multimodal Interaction* (Paris, France) *(ICMI '23)*. Association for Computing Machinery, New York, NY, USA, 594–603. https://doi.org/10.1145/3577190.3614131

[18] Shariff AM Faleel, Michael Gammon, Kevin Fan, Da-Yuan Huang, Wei Li, and Pourang Irani. 2021. HPUI: Hand Proximate User Interfaces for One-Handed Interactions on Head Mounted Displays. *IEEE Transactions on Visualization and Computer Graphics* 27, 11 (2021), 4215–4225. https://doi.org/10.1109/TVCG.2021.3106493

[19] Jacqui Fashimpaur, Amy Karlson, Tanya R. Jonker, Hrvoje Benko, and Aakar Gupta. 2023. Investigating Wrist Deflection Scrolling Techniques for Extended Reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 386, 16 pages. https://doi.org/10.1145/3544548.3580870

[20] Stephen Fitchett and Andy Cockburn. 2009. Evaluating reading and analysis tasks on mobile devices: a case study of tilt and flick scrolling. In *Proceedings of the 21st Annual Conference of the Australian Computer-Human Interaction Special Interest Group: Design: Open 24/7* (Melbourne, Australia) *(OZCHI '09)*. Association for Computing Machinery, New York, NY, USA, 225–232. https://doi.org/10.1145/1738826.1738863

[21] Patrick Grady, Jeremy A. Collins, Chengcheng Tang, Christopher D. Twigg, Kunal Aneja, James Hays, and Charles C. Kemp. 2024. PressureVision++: Estimating Fingertip Pressure From Diverse RGB Images. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 8698–8708.

[22] Juan David Hincapié-Ramos, Xiang Guo, Paymahn Moghadasian, and Pourang Irani. 2014. Consumed endurance: a metric to quantify arm fatigue of mid-air interactions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) *(CHI '14)*. Association for Computing Machinery, New York, NY, USA, 1063–1072. https://doi.org/10.1145/2556288.2557130

[23] Ken Hinckley, Edward Cutrell, Steve Bathiche, and Tim Muss. 2002. Quantitative analysis of scrolling techniques. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Minneapolis, Minnesota, USA) *(CHI '02)*. Association for Computing Machinery, New York, NY, USA, 65–72. https://doi.org/10.1145/503376.503389

[24] Nikhita Joshi, Parastoo Abtahi, Raj Sodhi, Nitzan Bartov, Jackson Rushing, Christopher Collins, Daniel Vogel, and Michael Glueck. 2023. Transferable Microgestures Across Hand Posture and Location Constraints: Leveraging the Middle, Ring, and Pinky Fingers. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3586183.3606713

[25] F. Jung, T. Dorszewski, R. Seibt, J.D. Glenday, D.F.B. Haeufle, and B. Steinhilber. 2023. Wrist position affects muscle fatigue during isometric contractions of wrist flexors: An exploratory study. *International Journal of Industrial Ergonomics* 98 (2023), 103507. https://doi.org/10.1016/j.ergon.2023.103507

[26] Daehwa Kim, Eric Whitmire, Roger Boldu, Wolf Kienzle, and Hrvoje Benko. 2024. SoundScroll: Robust Finger Slide Detection Using Friction Sound and Wrist-Worn Microphones. In *Proceedings of the 2024 ACM International Symposium on Wearable Computers* (Melbourne VIC, Australia) *(ISWC '24)*. Association for Computing Machinery, New York, NY, USA, 63–70. https://doi.org/10.1145/3675095.3676614

[27] Jina Kim, Yang Zhang, and Sang Ho Yoon. 2025. T2IRay: Design of Thumb-to-Index based Indirect Pointing for Continuous and Robust AR/VR Input. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 854, 16 pages. https://doi.org/10.1145/3706598.3713442

[28] Won Kim, F. Tendick, S. Ellis, and L. Stark. 1987. A comparison of position and rate control for telemanipulations with consideration of manipulator system dynamics. *IEEE Journal on Robotics and Automation* 3, 5 (1987), 426–436. https://doi.org/10.1109/JRA.1987.1087117

[29] Kenrick Kin, Chengde Wan, Ken Koh, Andrei Marin, Necati Cihan Camgöz, Yubo Zhang, Yujun Cai, Fedor Kovalev, Moshe Ben-Zacharia, Shannon Hoople, Marcos Nunes-Ueno, Mariel Sanchez-Rodriguez, Ayush Bhargava, Robert Wang, Eric Sauser, and Shugao Ma. 2024. STMG: A Machine Learning Microgesture Recognition System for Supporting Thumb-Based VR/AR Input. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 753, 15 pages. https://doi.org/10.1145/3613904.3642702

[30] Vincent Lambert, Adrien Chaffangeon Caillet, Alix Goguey, Sylvain Malacria, and Laurence Nigay. 2023. Studying the Visual Representation of Microgestures. *Proc. ACM Hum.-Comput. Interact.* 7, MHCI, Article 225 (Sept. 2023), 36 pages. https://doi.org/10.1145/3604272

[31] Vincent Lambert, Alix Goguey, Sylvain Malacria, and Laurence Nigay. 2024. Studying the Simultaneous Visual Representation of Microgestures. *Proc. ACM Hum.-Comput. Interact.* 8, MHCI, Article 276 (Sept. 2024), 34 pages. https://doi.org/10.1145/3676523

[32] Sylvain Malacria, Jonathan Aceituno, Philip Quinn, Géry Casiez, Andy Cockburn, and Nicolas Roussel. 2015. Push-Edge and Slide-Edge: Scrolling by Pushing Against the Viewport Edge. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) *(CHI '15)*. Association for Computing Machinery, New York, NY, USA, 2773–2776. https://doi.org/10.1145/2702123.2702132

[33] Sylvain Malacria, Alix Goguey, Gilles Bailly, and Géry Casiez. 2016. Multi-touch trackpads in the wild. In *Actes de La 28ième Conference Francophone Sur l'Interaction Homme-Machine* (Fribourg, Switzerland) *(IHM '16)*. Association for Computing Machinery, New York, NY, USA, 19–24. https://doi.org/10.1145/3004107.3004113

[34] Sylvain Malacria, Eric Lecolinet, and Yves Guiard. 2010. Clutch-free panning and integrated pan-zoom control on touch-sensitive surfaces: the cyclostar approach. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) *(CHI '10)*. Association for Computing Machinery, New York, NY, USA, 2615–2624. https://doi.org/10.1145/1753326.1753724

[35] M.J. Massimino, T.B. Sheridan, and J.B. Roseborough. 1989. One handed tracking in six degrees of freedom. In *Conference Proceedings., IEEE International Conference on Systems, Man and Cybernetics*. 498–503 vol.2. https://doi.org/10.1109/ICSMC.1989.71346

[36] Tomer Moscovich and John F. Hughes. 2004. Navigating documents with the virtual scroll ring. In *Proceedings of the 17th Annual ACM Symposium on User Interface Software and Technology* (Santa Fe, NM, USA) *(UIST '04)*. Association for Computing Machinery, New York, NY, USA, 57–60. https://doi.org/10.1145/1029632.1029642

[37] Robert Neßelrath, Mohammad Mehdi Moniri, and Michael Feld. 2016. Combining Speech, Gaze, and Micro-gestures for the Multimodal Control of In-Car Functions. In *2016 12th International Conference on Intelligent Environments (IE)*. 190–193. https://doi.org/10.1109/IE.2016.42

[38] Philip Quinn and Andy Cockburn. 2009. Zoofing! faster list selections with pressure-zoom-flick-scrolling. In *Proceedings of the 21st Annual Conference of the Australian Computer-Human Interaction Special Interest Group: Design: Open 24/7* (Melbourne, Australia) *(OZCHI '09)*. Association for Computing Machinery, New York, NY, USA, 185–192. https://doi.org/10.1145/1738826.1738856

[39] Philip Quinn, Andy Cockburn, Géry Casiez, Nicolas Roussel, and Carl Gutwin. 2012. Exposing and understanding scrolling transfer functions. In *Proceedings of the 25th Annual ACM Symposium on User Interface Software and Technology* (Cambridge, Massachusetts, USA) *(UIST '12)*. Association for Computing Machinery, New York, NY, USA, 341–350. https://doi.org/10.1145/2380116.2380161

[40] Philip Quinn, Sylvain Malacria, and Andy Cockburn. 2013. Touch scrolling transfer functions. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology* (St. Andrews, Scotland, United Kingdom) *(UIST '13)*. Association for Computing Machinery, New York, NY, USA, 61–70. https://doi.org/10.1145/2501988.2501995

[41] Jeff Sauro and James R. Lewis. 2010. Average task times in usability tests: what to report?. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) *(CHI '10)*. Association for Computing Machinery, New York, NY, USA, 2347–2350. https://doi.org/10.1145/1753326.1753679

[42] Selina Sharmin, Oleg Špakov, and Kari-Jouko Räihä. 2013. Reading on-screen text with gaze-based auto-scrolling. In *Proceedings of the 2013 Conference on Eye Tracking South Africa* (Cape Town, South Africa) *(ETSA '13)*. Association for Computing Machinery, New York, NY, USA, 24–31. https://doi.org/10.1145/2509315.2509319

[43] Grham Smith, m. c. schraefel, and Patrick Baudisch. 2005. Curve dial: eyes-free parameter entry for GUIs. In *CHI '05 Extended Abstracts on Human Factors in Computing Systems* (Portland, OR, USA) *(CHI EA '05)*. Association for Computing Machinery, New York, NY, USA, 1146–1147. https://doi.org/10.1145/1056808.1056855

[44] Mohamed Soliman, Franziska Mueller, Lena Hegemann, Joan Sol Roo, Christian Theobalt, and Jürgen Steimle. 2018. FingerInput: Capturing Expressive Single-Hand Thumb-to-Finger Microgestures. In *Proceedings of the 2018 ACM International Conference on Interactive Surfaces and Spaces* (Tokyo, Japan) *(ISS '18)*. Association for Computing Machinery, New York, NY, USA, 177–187. https://doi.org/10.1145/3279778.3279799

[45] Yanke Tan, Sang Ho Yoon, and Karthik Ramani. 2017. BikeGesture: User Elicitation and Performance of Micro Hand Gesture as Input for Cycling. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) *(CHI EA '17)*. Association for Computing Machinery, New York, NY, USA, 2147–2154. https://doi.org/10.1145/3027063.3053075

[46] Jérémy Wambecke, Alix Goguey, Laurence Nigay, Lauren Dargent, Daniel Hauret, Stéphanie Lafon, and Jean-Samuel Louis de Visme. 2021. M[eye]cro: Eye-gaze+Microgestures for Multitasking and Interruptions. *Proc. ACM Hum.-Comput. Interact.* 5, EICS, Article 210 (may 2021), 22 pages. https://doi.org/10.1145/3461732

[47] Eric Whitmire, Mohit Jain, Divye Jain, Greg Nelson, Ravi Karkar, Shwetak Patel, and Mayank Goel. 2017. DigiTouch: Reconfigurable Thumb-to-Finger Input and Text Entry on Head-mounted Displays. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 3, Article 113 (Sept. 2017), 21 pages. https://doi.org/10.1145/3130978

[48] Pawel Woźniak, Morten Fjeld, and Shengdong Zhao. 2014. Limiting trial and error: introducing a systematic approach to designing clutching. In *Proceedings of the Second International Symposium of Chinese CHI* (Toronto, Ontario, Canada) *(Chinese CHI '14)*. Association for Computing Machinery, New York, NY, USA, 35–39. https://doi.org/10.1145/2592235.2592241

[49] Shumin Zhai. 1995. Human performance in six degree of freedom input control. (01 1995).

[50] Shumin Zhai, Barton A Smith, and Ted Selker. 1997. Improving Browsing Performance: A study of four input devices for scrolling and pointing tasks. In *Human-Computer Interaction INTERACT'97: IFIP TC13 International Conference on Human-Computer Interaction, 14th–18th July 1997, Sydney, Australia*. Springer, 286–293.

[51] Xiaolei Zhou. 2014. Comparative Study on Cursor Position Controlled by Pen Pressure and Pen Tilt. In *2014 Seventh International Symposium on Computational Intelligence and Design*, Vol. 2. 375–378. https://doi.org/10.1109/ISCID.2014.151

[52] Quentin Zoppis, Sergi Pujades, Laurence Nigay, and François Bérard. 2025. Efficient Finger Model and Accurate Tracking for Hover-and-Touch, Mid-air and Microgesture Interaction. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. Association for Computing Machinery, New York, NY, USA. https://hal.science/hal-05247942v1/document